Supplemental Material
Vacancy-Induced Dzyaloshinskii–Moriya Interaction and Spin Canting in Monolayer Fe_3GeTe_2

Nikhil Kodali, Biplab Sanyal, and Phani Motamarri

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S1 Electronic-structure calculations

This section lists the computational settings summarized in the Methods section of the main text. We solve the noncollinear Kohn–Sham equations [1] of density functional theory (DFT) with spin–orbit coupling (SOC), using the finite-element code DFT-FE [2–4]. The exchange-correlation (XC) functional is the Perdew–Burke–Ernzerhof (PBE) generalized-gradient approximation [5], supplemented by D4 dispersion corrections that include the three-body term [6, 7]. We use fully relativistic optimized norm-conserving Vanderbilt (ONCV) pseudopotentials from the PseudoDojo stringent set [8, 9], with 16, 22, and 24 valence electrons for Fe, Ge, and Te. Our test calculations at lower settings, some of which used pseudopotentials with 16, 14, and 16 valence electrons, enter only the comparisons in Sec. S5.

The supercell is orthorhombic, with cell vectors of lengths 24.0683375, 20.8633296, and 19.8014567 Å, and the boundary conditions are periodic along x and y and nonperiodic along z. The pristine monolayer Fe_3GeTe_2 (FGT) cell contains 108 Fe, 36 Ge, and 72 Te atoms, with the Fe atoms on two sublattices, Fe(1) in the two outer Fe planes and Fe(2) in the central Fe–Ge plane. Removing one Fe atom gives $Fe_{2.9722}GeTe_2$, in which 0.926% of all Fe sites are vacant, and lowers the total occupation from 4248 electrons in the pristine cell to 4232 in each single-vacancy cell. For an Fe(2) vacancy, 1 of the 36 Fe(2) sites (2.78%) is vacant. All results refer to this single supercell size, without extrapolation to zero vacancy concentration.

The production calculations use seventh-order finite elements for the wave functions and the electrostatic potential, fourteenth-order interpolation of the density, and a density quadrature of order 15, on an adaptively generated mesh whose base element size of 6.5 bohr is refined to 0.85 bohr within 5 bohr of each atom. The nonlocal pseudopotential and the local potential are integrated with a Gauss quadrature of order 40 in each element. The Brillouin-zone sums include all 16 points of the $4 \times 4 \times 1$ Monkhorst–Pack grid [10], without reduction by spatial or time-reversal symmetry, and the Fermi–Dirac occupations use an electronic temperature of 50 K.

The self-consistent-field (SCF) iterations use Anderson mixing [11] with Resta preconditioning [12], with a mixing parameter of 0.3, a history length of 50, and spin-mixing enhancement factors of 3 for the longitudinal and 6 for the transverse magnetization, and they are converged to a density-residual tolerance of 2×10^{-5}. An option of the mixing that accelerates a uniform rotation of the magnetization was enabled, but its gain remained equal to one in all iterations of all calculations. Residual-based Chebyshev filtered subspace iteration [13] is performed with the mixed-precision implementation of Ref. [14], in which the ground-state (GS) calculations use TensorFloat-32 (TF32) arithmetic and bfloat16 (BF16) communication in the Chebyshev filtering. For the converged Hamiltonian, we then recompute the wave functions in non-self-consistent (NSCF) calculations in single precision (FP32), without reduced-precision communication.

Atomic positions were relaxed with SOC at fixed cell vectors. The GS calculations for pristine FGT, the Fe(1) vacancy, and the Fe(2) vacancy converge in 47, 51, and 49 SCF iterations, with largest residual forces of 0.00524, 0.01487, and 0.01299 eV/Å, respectively. The Fe(1)-vacancy cell is higher in energy than the Fe(2)-vacancy cell by 0.4073 eV in internal energy and 0.4078 eV in electronic free energy. Because both cells have the same composition, these values are relative energies of the two vacancy sites rather than absolute vacancy formation energies.

S1.1 Relaxation around the vacancies

Figure S1 shows, plane by plane, how the atoms move when a vacancy is created, measured from the matched positions of the relaxed pristine cell after removing the small rigid shift of the whole cell. The two vacancies relax differently. Around the Fe(2) vacancy, the three Ge atoms of the central plane move 0.30–0.33 Å toward the vacant site, by far the largest displacement in either cell, and the nearest central-plane Fe(2) atoms at 4.0 Å follow with 0.09–0.10 Å, mainly toward the vacancy. The Te atoms directly above and below the vacancy move 0.06 Å toward the central plane, whereas the six nearest Fe(1) atoms, the most canted atoms of this cell, move by at most 0.013 Å. Around the Fe(1) vacancy, the displacements are smaller and mostly vertical. The Fe(1) atom directly below the vacancy moves 0.10 Å toward it, the three nearest Ge atoms move 0.09 Å upward and slightly inward, the nearest upper Te atoms move 0.08 Å downward and inward, and the three nearest Fe(2) atoms move 0.05 Å away from the vacancy plane. Beyond 9 Å the root-mean-square displacement is 0.005 Å for the Fe(1) vacancy and 0.014 Å for the Fe(2) vacancy, which shows that the relaxation of the Fe(2)-vacancy cell extends further. All maps and distances in the main text use the pristine positions, so these displacements do not enter the plotted geometry.

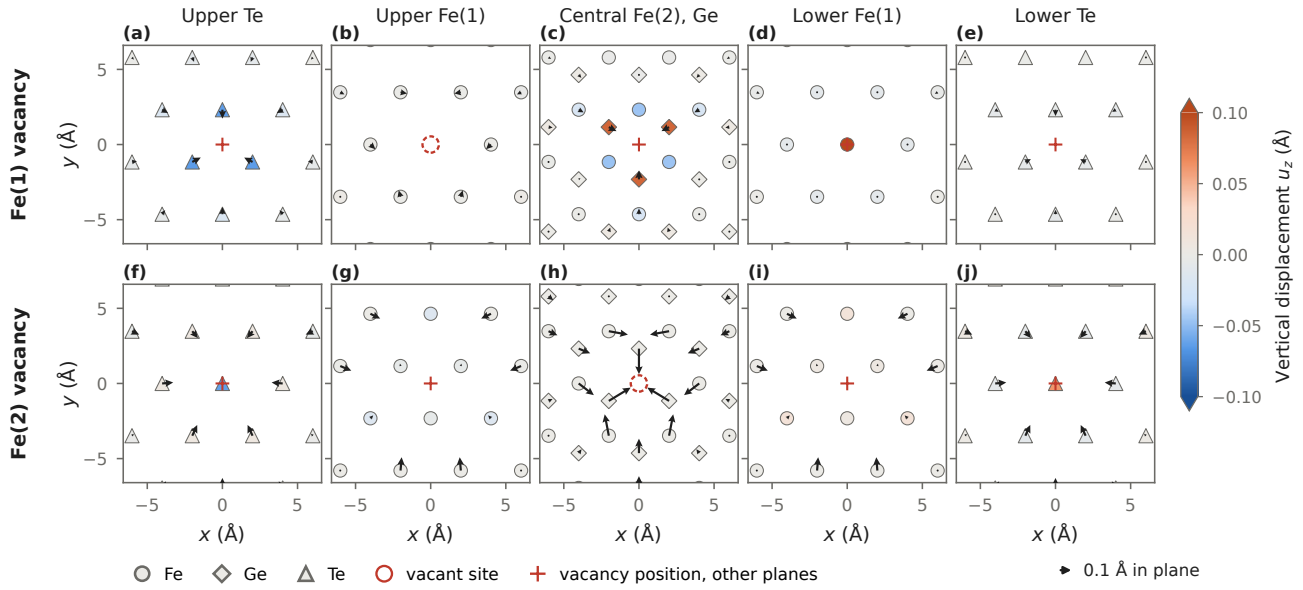


Figure S1: Relaxation displacements **u** around the Fe(1) vacancy (a–e) and the Fe(2) vacancy (f–j), relative to the relaxed pristine cell, shown plane by plane from the upper Te plane to the lower Te plane. Atoms are drawn at their pristine positions relative to the vacancy. Arrows give the in-plane displacement, magnified 14 times, and marker colors give the vertical displacement u_z (positive upward). In-plane displacements below 0.004 Å are not drawn. The three Ge arrows in (h), with true lengths of 0.30–0.33 Å, are shortened to fit. The dashed circle marks the vacant site and the cross its in-plane position in the other planes.

S2 Atomic moments, radial components, and layer symmetry

This section defines the atomic moments, canting angles, and radial and layer-resolved measures used in the main text and tests how they depend on the integration window. Atomic moments, in μ_B, are integrals of the occupied spin density over a tapered sphere, with the convention $\mathbf{m} = 2\mathbf{S}$, $\mathbf{S} = \boldsymbol{\sigma}/2$, and $\hbar = 1$. For inner and outer radii a and b, the window is $W = 1$ for $r \le a$ and $W = 0$ for $r \ge b$, and with $x = (r-a)/(b-a)$ we use two tapers,

$$W_{\mathrm{cosine}}(x) = \frac{1+\cos(\pi x)}{2}, \qquad W_{\mathrm{smooth}}(x) = 1 - \frac{1}{1+\exp[1/x - 1/(1-x)]}. \tag{S1}$$

Unless stated otherwise, atomic moments use the cosine taper with $(a, b) = (1.8, 2.1)$ bohr, which we call the cosine window. The windows of different atoms do not overlap, and Table S1 shows how the maximum canting angles change across six windows.

All atomic moments in this work are spin moments, $\boldsymbol{\mu}_A = \int W_A\, \mathbf{m}\, d\mathbf{r}$, with $\mathbf{m}$ the spin magnetization density. The orbital moments in the same windows are about 0.055 μ_B per Fe atom and, in the pristine cell, nearly parallel to the spin moments (largest angle 0.1°). Near the vacancies, however, they cant more strongly than the spin moments, by up to 5° (Fe(2) vacancy) and 13° (Fe(1) vacancy). For atoms with spin canting angles above 0.3°, the ratio of the canting angle of the total (spin plus orbital) moment to that of the spin moment is 1.21–1.32 for the Fe(2) vacancy and 0.96–1.31 for the Fe(1) vacancy, and the largest total-moment canting angles are 0.54° and 1.17°, respectively.

The canting angle of atom A is measured from the direction of the summed Fe moment,

$$\theta_A = \mathrm{atan2}(|\boldsymbol{\mu}_A \times \widehat{\mathbf{M}}_{\mathrm{Fe}}|, \boldsymbol{\mu}_A \cdot \widehat{\mathbf{M}}_{\mathrm{Fe}}), \qquad \mathbf{M}_{\mathrm{Fe}} = \sum_{A \in \mathrm{Fe}} \boldsymbol{\mu}_A. \tag{S2}$$

For the atomic moments, a common rotation of all atomic vectors of a calculation takes $\widehat{\mathbf{M}}_{\mathrm{Fe}}$ to z, whereas the vacuum spin polarization and the SOC torques are given in the laboratory Cartesian axes. The summed Fe moment is tilted from the laboratory z axis by 0.0201° in the Fe(1)-vacancy calculation and 0.0057° in the Fe(2)-vacancy calculation.

To compare a vacancy cell with the pristine cell atom by atom, we match each atom of the vacancy cell one to one, by species and nearest minimum-image distance, to an atom of the pristine cell, which we call the corresponding pristine site. All distances and radial directions are taken from these pristine positions relative to the removed atom. We define the radial fraction of a set of atoms as $\sum_A (\boldsymbol{\mu}_{A,\perp} \cdot \mathbf{e}_{r,A})^2 / \sum_A |\boldsymbol{\mu}_{A,\perp}|^2$, without density weighting. For the six upper and six lower Fe atoms within a three-dimensional distance $r < 5$ Å of the Fe(2) vacancy, excluding atoms on the vacancy axis, this fraction is 0.9877 in the upper plane and 0.9887 in the lower plane.

To compare the two outer planes, we pair upper and lower Fe(1) atoms at the same in-plane position. With transverse moments $\mathbf{u}_i$ and $\mathbf{l}_i$, the layer-antisymmetric fraction is

$$\eta_- = \frac{\sum_i |\mathbf{u}_i - \mathbf{l}_i|^2}{\sum_i |\mathbf{u}_i - \mathbf{l}_i|^2 + \sum_i |\mathbf{u}_i + \mathbf{l}_i|^2}. \tag{S3}$$

For the Fe(2) vacancy, $\eta_- = 0.9991$ over all 36 pairs, which means that the transverse moments of the two planes are opposite. For the Fe(1) vacancy, by contrast, $\eta_- = 0.0624$ over the 35 remaining pairs, and the transverse moments are nearly parallel. Because the Fe(2) vacancy retains the horizontal mirror, which maps an axial vector as $(m_x, m_y, m_z) \mapsto (-m_x, -m_y, m_z)$, symmetry requires $\eta_- = 1$ for this vacancy, and the computed value serves as a numerical check. To test whether the relaxed structures retain this mirror, we place the mirror plane at the mean z coordinate of the paired Fe atoms, which removes a rigid translation. The largest distance between an Fe position and the mirror image of its partner is then 2.36×10^{-4} Å for the Fe(2) vacancy and 0.0285 Å for the Fe(1) vacancy. The test

involves only the Fe pairs and is not a space-group classification of the relaxed cells. The mirror is present, to within 5×10^{-4} Å, in every recorded geometry of the relaxation of the Fe(2)-vacancy cell. The relaxation therefore does not test the stability of this cell against a displacement of Fe(2) atoms out of the central plane. Such a displacement was reported for Fe-deficient FGT [15] and would break the horizontal mirror.

Window	Taper	Pristine	Fe(1) vac.	Fe(2) vac.	Fe(2) vac. η_-
1.8,2.1	cosine	0.014073	0.948481	0.444107	0.999146
1.6,1.9	cosine	0.014003	0.944996	0.443830	0.999144
1.7,2	cosine	0.014036	0.946599	0.443983	0.999145
1.2,2.1	smooth	0.013978	0.943993	0.443624	0.999143
1.2,1.9	smooth	0.013950	0.942813	0.443413	0.999142
1.2,2	smooth	0.013964	0.943362	0.443524	0.999143

Table S1: Maximum Fe canting angle, in degrees, for six integration windows. Window radii a, b are in bohr. The final column is η_- for the Fe(2) vacancy. For the Fe(1) vacancy, η_- ranges from 0.06120 to 0.06240 over the same windows. All rows use the same self-consistent states, and only the definition of the atomic moment changes.

The canting extends beyond the region shown in Fig. 2 of the main text, as Fig. 3 of the main text, which includes all Fe atoms, shows. Beyond 7 Å from the vacancy, the root-mean-square (RMS) Fe canting angle is approximately 0.184° for the Fe(1) vacancy and 0.075° for the Fe(2) vacancy. Because of these tails, the computed textures cannot be identified with those of an isolated vacancy.

S2.1 Features of the textures required by symmetry

Several features of the canting patterns discussed in the main text follow from symmetry alone, and this subsection states and tests the corresponding constraints. With the magnetization along z, the magnetic symmetry operations about a vacancy are the horizontal mirror σ_h, if retained, and vertical mirrors combined with time reversal, $\sigma_v T$. The operation $\sigma_v T$ leaves m_z and the in-plane component lying in the mirror plane unchanged and reverses the in-plane component normal to it. Together, these operations impose three constraints. (i) For the Fe(2) vacancy, σ_h requires opposite transverse moments in the two outer planes and zero transverse moments on all central-plane sites. (ii) On a vertical mirror plane through the vacancy, the transverse moment is radial. (iii) Off such a plane, the tangential component is odd and the radial component even under the mirror. Both vacancy sites have a local threefold axis with three vertical mirrors. The rectangular periodic cell, however, breaks the threefold axis and retains exactly only one of the vertical mirrors, $x \to -x$ through the vacancy, for which the largest mismatch of mirrored atomic positions is 8×10^{-4} Å for the Fe(1) vacancy. The Fe(2) vacancy also retains σ_h.

To test these constraints, Table S2 compares Fe atoms on and off the exact mirror. For the Fe(2) vacancy, the largest central-plane canting angle is 0.0066°, below the pristine maximum. Radial components account for 99.98% of the sum of squared transverse moments on the exact mirror, for 98.4% off it within 5 Å (where the nearest Fe(1) neighbors lie on the approximate local mirrors), and for 71% off it over the whole cell. For the nearest shells alone, the radial share is 99.1% (six Fe(1) neighbors of the Fe(2) vacancy) and 99.6% (three Fe(2) neighbors of the Fe(1) vacancy). For the Fe(1) vacancy, the corresponding off-mirror value within 5 Å is 37%, and Table S3 lists the nearest Fe shells of this vacancy.

The argument for the horizontal mirror carries over to the bulk crystal, whose space group is $P6_3/mmc$ [16]. Its unit cell contains two layers, and the Fe(2) atoms occupy the Wyckoff position $2c$, with one site in each layer. If the two sites are occupied unequally and nothing else changes, the

screw axis and the inversion that relate the two layers are lost. The threefold axis, the vertical mirrors, the twofold axes in the central planes, and the mirror through the central plane of each layer remain. The space group is then $P\bar{6}m2$ and the point group is D_{3h}, for which no Lifshitz invariant exists. An unequal occupancy of the Fe(2) sites in alternate layers therefore allows no net DMI by itself. A polar structure requires a further distortion that removes the mirror through the central plane of each layer. Reference [17] reports occupancies of 0.86 and 0.92 for the two Fe(2) sites and attributes the loss of inversion symmetry mainly to this imbalance, which is consistent with $P\bar{6}m2$. The polar space group $P3m1$ of its refinement in addition requires the reported Fe atoms in the van der Waals gap or the reported displacements along z, which are below 0.05 Å.

Vacancy	Fe atoms	n	radial fraction	max $\lvert\mu_{\mathrm{tan}}\rvert$ (μ_B)
Fe(2)	on mirror, $r < 5$ Å	4	0.9998	0.0002
Fe(2)	off mirror, $r < 5$ Å	14	0.984	0.0019
Fe(2)	off mirror, all	99	0.709	0.0064
Fe(1)	on mirror, $r < 5$ Å	2	1.0000	0.00003
Fe(1)	off mirror, $r < 5$ Å	16	0.368	0.0316
Fe(1)	off mirror, all	99	0.490	0.0316

Table S2: Radial fraction $\sum(\boldsymbol{\mu}_\perp \cdot \mathbf{e}_r)^2 / \sum|\boldsymbol{\mu}_\perp|^2$ of the Fe transverse moments on and off the exact $x \to -x$ mirror through the vacancy, excluding atoms on the vacancy axis. Here r is the three-dimensional distance to the removed atom, and μ_{tan} the tangential component.

Plane	r (Å)	n	θ (°)	$\lvert\boldsymbol{\mu}\rvert$ (μ_B)	μ_{rad} (μ_B)	$\lvert\mu_{\mathrm{tan}}\rvert$ (μ_B)
central	2.62	3	0.79–0.95	1.23	−0.017 to −0.020	≤ 0.0014
upper	4.01	6	0.53–0.62	2.30	+0.006 to +0.013	0.018–0.023
central	4.79	3	0.37–0.45	1.47	+0.009 to +0.011	≤ 0.0004
lower	2.46	1	0.09	2.56	on axis	
lower	4.70	6	0.76–0.90	2.44	+0.019 to +0.025	0.026–0.032

Table S3: Nearest Fe shells of the Fe(1) vacancy, in the frame in which the summed Fe moment is along z. θ is the canting angle, and μ_{rad} and μ_{tan} are the radial and tangential transverse components. Positive radial components point away from the vacancy. Pristine moments are 2.43 (Fe(1)) and 1.50 μ_B (Fe(2)). Tangential components change sign across the $x \to -x$ mirror. For comparison, the six nearest Fe(1) neighbors of the Fe(2) vacancy (2.62 Å) have $\theta = 0.39$–$0.44°$, $|\boldsymbol{\mu}| = 2.11$ μ_B, radial components ∓ 0.014 to ∓ 0.016 μ_B (upper inward, lower outward), and tangential components ≤ 0.002 μ_B.

S2.2 Pairs of Fe(2) vacancies

To test how two vacancies in the same cell modify each other's effect on the local moments, we also computed two cells that each contain two Fe(2) vacancies, at in-plane minimum-image separations of 4.014 and 14.47 Å. Both cells have 214 atoms and the settings of the single-vacancy cells. The cell with the separated pair was computed with the later version of the code and the FP32 arithmetic that were used for the states magnetized along x (Sec. S6). We compare them with the sum of the single-vacancy changes. Let $\Delta\boldsymbol{\mu}_i^{(1)}(\mathbf{R}_v)$ be the change of the moment of atom i caused by a single Fe(2) vacancy at $\mathbf{R}_v$, constructed by translating the single-vacancy result on the pristine lattice. The additive reference is then

$$\boldsymbol{\mu}_i^{\mathrm{add}} = \boldsymbol{\mu}_i^{(0)} + \Delta\boldsymbol{\mu}_i^{(1)}(\mathbf{R}_a) + \Delta\boldsymbol{\mu}_i^{(1)}(\mathbf{R}_b). \tag{S4}$$

The comparison uses the cosine window and the laboratory axes common to all calculations. We define the deviation from additivity as the change in moment magnitude $d_i = |\boldsymbol{\mu}_i^{(2)}| - |\boldsymbol{\mu}_i^{\mathrm{add}}|$ and take its RMS value over all 106 remaining Fe atoms. Aligning the summed Fe moment of each calculation with z changes the deviations by less than 10^{-6} μ_B. Table S4 summarizes the results.

Pair	R (Å)	ΔF_2 (eV)	RMS d_i (μ_B)	d_{ext} (μ_B)
Close	4.0143	0.31443	0.03049	-0.20876
Separated	14.4703	0.02867	0.00771	-0.02134

Table S4: Pairs of Fe(2) vacancies. ΔF_2 is $F_{2v} + F_0 - 2F_{1v}$ in the periodic cell, and d_{ext} is the deviation from additivity with the largest magnitude, including its sign. Energies are electronic free energies at the electronic temperature of 50 K.

The close pair is higher in energy than the separated pair by 0.2858 eV in electronic free energy and 0.2846 eV in internal energy. Its largest deviation from additivity occurs at an Fe atom whose moment is 1.6285 μ_B, compared with the additive value 1.8372 μ_B. With a smaller cosine window (outer radius 1.9 bohr), the largest deviations are about -0.201 and -0.0206 μ_B. These deviations include the effects of both structural relaxation and electronic response. The separated pair serves as a reference within this periodic cell but may retain a residual interaction, and its small excess energy is therefore not a converged interaction energy. The transverse moments of the close pair are also less mirror-antisymmetric ($\eta_- = 0.956$) than those of the separated pair (0.99997) and the single Fe(2) vacancy (0.9991), although its Fe positions are mirrored to within 3×10^{-4} Å. We do not use the pairs as a test of the canting.

S3 Site- and orbital-resolved spin spectra

To examine how the transverse moments of the nearest neighbors are distributed over the occupied states, this section resolves the occupied spin density at the Fe sites nearest each vacancy by energy and by orbital. Energy-dependent spectral spin directions in noncollinear magnets have been reported before [18]. The energy- and orbital-resolved quantities describe the spin density of the self-consistent states and are not canting energies or exchange parameters.

S3.1 Definitions and atomic groups

For an atomic window W_A, the site-resolved spectral charge and spin densities are

$$n_A(E) = \sum_{n\mathbf{k}} w_{\mathbf{k}} \langle u_{n\mathbf{k}} | W_A | u_{n\mathbf{k}} \rangle \delta(E - \varepsilon_{n\mathbf{k}} + E_F), \tag{S5}$$

$$\mathbf{m}_A(E) = \sum_{n\mathbf{k}} w_{\mathbf{k}} \langle u_{n\mathbf{k}} | W_A \boldsymbol{\sigma} | u_{n\mathbf{k}} \rangle \delta(E - \varepsilon_{n\mathbf{k}} + E_F). \tag{S6}$$

The corresponding spin-density matrix is $[n_A(E)1 + \mathbf{m}_A(E)\cdot\boldsymbol{\sigma}]/2$. The two spin-projected local densities of states (LDOS), $g_{A,\pm} = (n_A \pm m_{z,A})/2$, are projections onto the direction of the summed Fe moment, since spin is not a conserved quantum number in the presence of SOC. The transverse components carry information that these two projections omit. As for the atomic texture, the radial direction points from the in-plane position of the vacancy toward the corresponding pristine site, so negative radial spin components point inward.

The site-resolved spectra use the 1.2–2.1-bohr smooth window, the same window as the orbital projections of Sec. S3.4, whereas the atomic canting angles use the cosine window. The integrated site moments in this section are likewise computed with the smooth window. The spectra are nevertheless

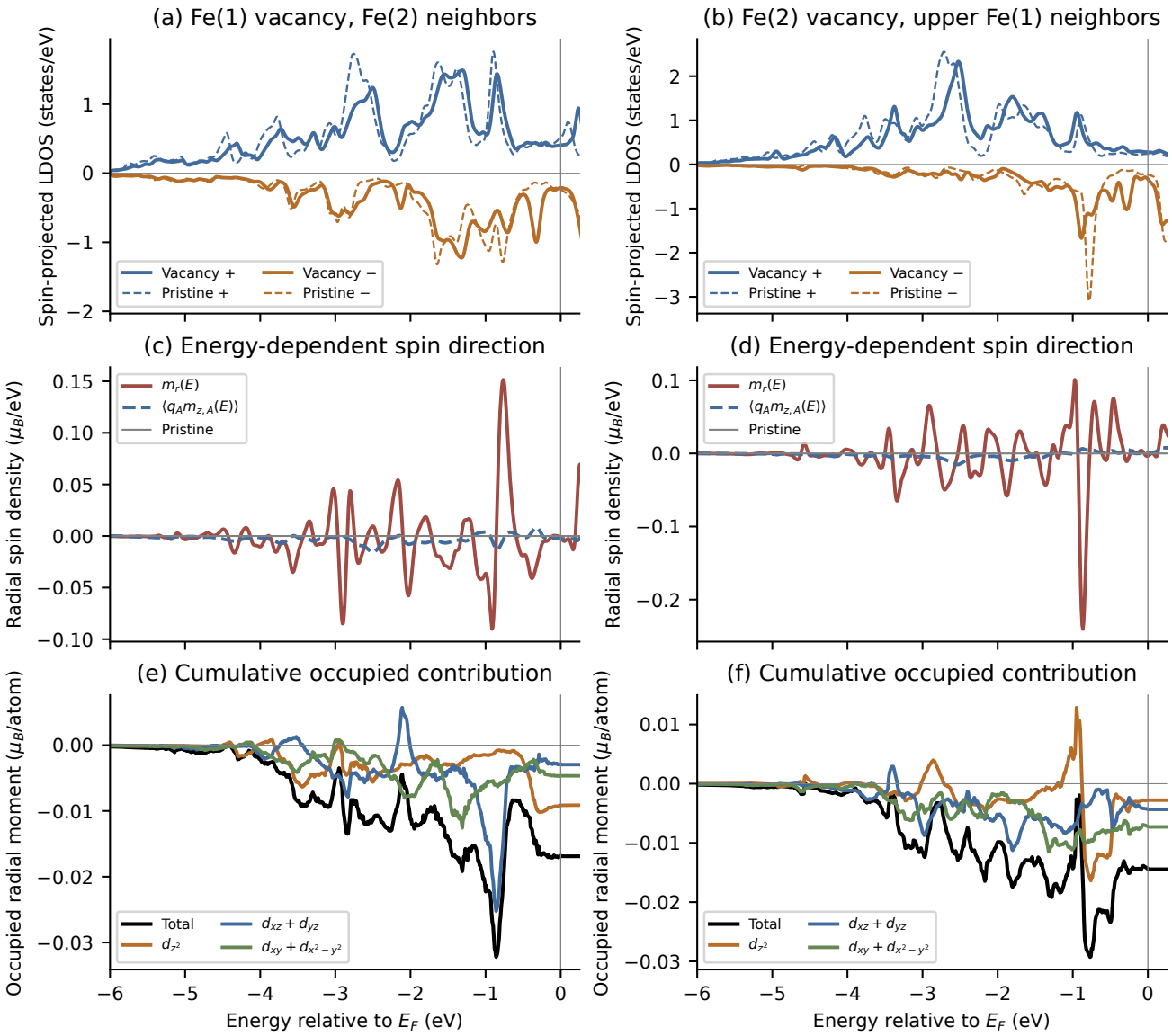


Figure S2: Site-resolved spin spectra and occupied orbital contributions. The two columns average the three nearest central-plane Fe(2) neighbors of the Fe(1) vacancy and the three nearest upper-plane Fe(1) neighbors of the Fe(2) vacancy. The top row shows the spin-projected local density of states (LDOS) of the site, vacancy solid and pristine dashed, with the minus projection plotted negatively. The middle row shows the radial spin density and the common-axis construction of Sec. S3.3, and the bottom row the cumulative occupied radial contributions, including all states below −6 eV.

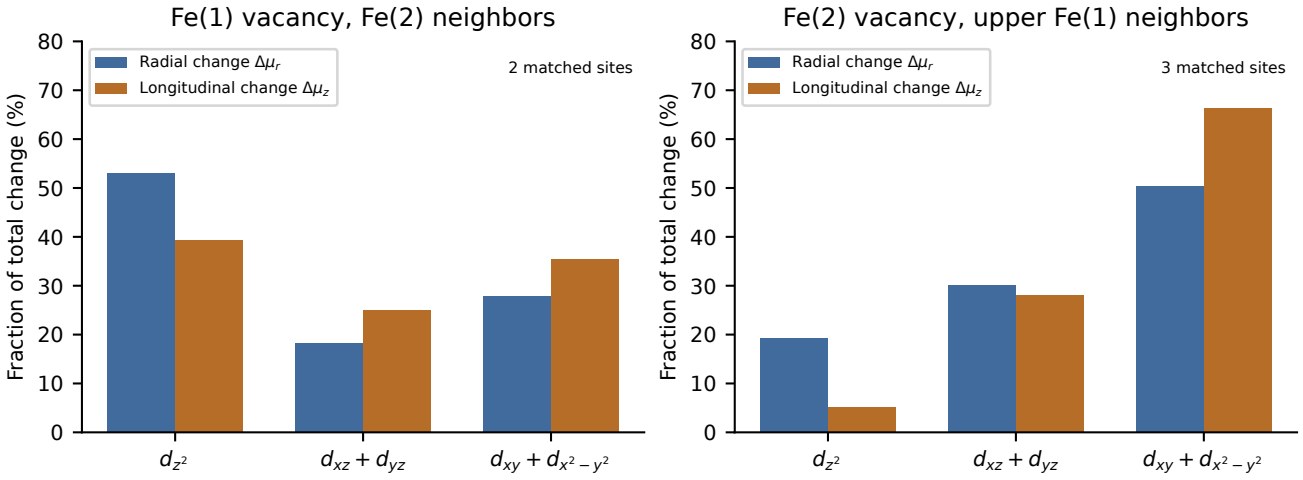


Figure S3: Occupied-orbital contributions to the spin changes relative to the corresponding pristine sites. Each bar is the share of an orbital group in the radial or longitudinal spin change. The Fe(1)-vacancy Fe(2) comparison uses the two sites whose corresponding pristine sites have orbital projections. The Fe(2)-vacancy upper Fe(1) comparison uses three. All projections use the 1.2–2.1-bohr smooth window.

expressed in the frame of the atomic texture, because every spin spectrum of a calculation is rotated by the rotation that takes the summed Fe moment, computed with the cosine window, to z.

We analyze two groups of three sites, all 2.62 Å from the removed atom. These are the three central-plane Fe(2) neighbors of the Fe(1) vacancy and the three upper-plane Fe(1) neighbors of the Fe(2) vacancy. For the orbital projections, we also use the three lower-plane Fe(1) neighbors of the Fe(2) vacancy, at the same distance, to test the response of the opposite plane. In each group, one site lies on the $x \to -x$ mirror and two lie off it, and Table S9 gives their in-plane positions.

The site LDOS averages include all three sites of each group, in both the vacancy and the pristine cell, and orbital projections are computed at all three sites of each vacancy group. For the Fe(1) vacancy, however, only two of the three corresponding pristine sites have orbital projections, and the orbital changes of that group are averaged over these two sites. The single-site orbital projected densities of states (PDOS) in Fig. S4 are for the site with the largest canting angle (cosine window) near each vacancy, which is the central-plane Fe(2) neighbor on the mirror, at $(x, y) = (0.00, +2.32)$ Å relative to the vacancy, for the Fe(1) vacancy and the upper-plane Fe(1) neighbor at $(+2.01, +1.16)$ Å for the Fe(2) vacancy. Table S5 lists the group-averaged moments.

Group	Pristine μ_z	Vacancy μ_z	Vacancy μ_r	Max. $\lvert\mu_r(E_c)\rvert$
Fe(1) vacancy, Fe(2)	1.406978	1.154672	-0.016889	0.032205
Fe(2) vacancy, upper Fe(1)	2.264472	1.965669	-0.014455	0.029270

Table S5: Smooth-window moments of the three-site groups, averaged over the group, in μ_B. The last column is the largest magnitude of the cumulative occupied radial moment for upper integration limits E_c between -6 and $+1$ eV, including all states below -6 eV. Because it exceeds the final value, contributions from different energy ranges partly cancel.

S3.2 Energy sums and numerical checks

This subsection describes how the spectra are accumulated and checks them against the directly integrated moments. The spectra sum the computed eigenstates with their k-point weights, and additional computed unoccupied states lie above the displayed energy range. At each k point, every group of degenerate eigenstates is assigned its mean eigenvalue, so that a bin boundary cannot split the basis-dependent contributions within the group, while the occupations of the individual states are retained. Spectral densities use 10-meV bins from -6 to $+1$ eV, with all occupied states below -6 eV included as a cumulative offset, whereas the full occupied sum extends over all computed states without this cutoff. Within the displayed interval, the largest shift of an eigenvalue to its group mean is 3.37×10^{-5} eV, well below the 10-meV bin width, and sums over the unshifted eigenvalues reproduce the $[-0.5, 0)$-eV contributions in Table S7 to numerical precision.

The density of states (DOS) curves in the spectral figures are smoothed with

$$K_w(E) = \frac{\exp[-(E/w)^2]}{\sqrt{\pi} w}, \qquad w = 0.05 \text{ eV}. \tag{S7}$$

The cumulative moment curves instead sum the occupation-weighted, unbroadened bins, so the integrated moments are independent of the broadening. The occupations are those of the 50-K SCF state and are unaffected by the smoothing.

System	Spectral vs. direct	Orbital sum rule	Orbital spin vs. site	Energy partition
Pristine	7.29×10^{-12}	1.78×10^{-14}	1.47×10^{-6}	1.60×10^{-14}
Fe(1) vac.	1.05×10^{-11}	3.20×10^{-14}	2.13×10^{-6}	1.78×10^{-14}
Fe(2) vac.	7.49×10^{-12}	3.55×10^{-14}	1.18×10^{-6}	1.60×10^{-14}

Table S6: Maximum absolute discrepancies of the spectral sums. Spectral vs. direct compares the occupied charge and spin integrals of the site spectra with the values computed directly in real space with the smooth and cosine windows. Orbital sum rule is the discrepancy in the sum rules of the processed orbital weights. Orbital spin vs. site compares the separately integrated spin within the window from the orbital projection with the site spin integral. Energy partition compares the sum of the offset below -6 eV and the binned contributions with the full sum. Charges are in electrons. Spin integrals are dimensionless, before conversion to μ_B.

Because the orbital projections use an atom-centered quadrature that is separate from the real-space site integration, the two need not agree exactly. The occupied spin components within the window from the two quadratures differ by at most 2.13×10^{-6} across the three systems, and the charges by up to 1.79×10^{-5} (Table S6). We report both integrals as computed, without substituting one for the other.

S3.3 Test of a common local spin axis

The spectra of Fig. S2 allow a direct test of whether the spin density of each site has a single axis at all energies. If it does, the spectrum can be written $\mathbf{m}_A(E) = s_A(E)\hat{\boldsymbol{\mu}}_A$, where $s_A(E)$ may take either sign, and it then obeys $m_{r,A}(E) = q_A m_{z,A}(E)$, with $q_A = \mu_{r,A}/\mu_{z,A}$. We construct this reference from the integrated moment and the computed longitudinal spectrum of each site, without an additional DFT calculation, and apply the individual ratios q_A before averaging over the group.

Let $d_r(E)$ be the group average of $m_{r,A}(E) - q_A m_{z,A}(E)$. Its full occupied integral vanishes by construction, and a nonzero $d_r(E)$ shows that the spectral spin direction varies with energy. For all three broadening widths in Table S8, the integrated residual $|d_r|$ is of the same magnitude as the integrated radial spectrum $|m_r|$.

Energy interval (eV)	Fe(1) vacancy Fe(2)	Fe(2) vacancy upper Fe(1)
$(-\infty, -6)$	-0.0002127	-0.0001775
$[-6, -3)$	-0.0047061	-0.0113681
$[-3, -1)$	-0.0173484	-0.0027863
$[-1, -0.5)$	0.0122708	-0.0090946
$[-0.5, 0)$	-0.0068172	0.0089910
$[0, 1)$	-0.0000759	-0.0000198
Full occupied sum	-0.0168894	-0.0144554

Table S7: Occupied radial-moment contributions, in μ_B per atom. The first row contains every occupied state below -6 eV, including semicore states in the stringent valence spaces. The small positive-energy contribution arises from the 50-K occupations. Contributions above $+1$ eV are negligible at this precision.

w (eV)	Fe(1) vac. $\int \lvert m_r \rvert$	Fe(1) vac. $\int \lvert d_r \rvert$	Fe(2) vac. $\int \lvert m_r \rvert$	Fe(2) vac. $\int \lvert d_r \rvert$
0.025	0.096498	0.097402	0.113985	0.114275
0.050	0.082881	0.083682	0.092906	0.092962
0.100	0.057293	0.057893	0.061246	0.060337

Table S8: Broadening dependence of the common-axis test. The integrals of the absolute group-averaged radial spin spectrum and of its common-axis residual are taken over $[-3, -0.05]$ eV, in μ_B per atom.

S3.4 Orbital projections

Each spinor is projected onto real spherical harmonics centered at the selected atom. With $a_{\ell m}(r)$ the radial spinor component for harmonic ℓm,

$$n_{\ell m} = \int r^2 W(r) a^\dagger_{\ell m} a_{\ell m}\, dr, \qquad \mathbf{m}_{\ell m} = \int r^2 W(r) a^\dagger_{\ell m} \boldsymbol{\sigma} a_{\ell m}\, dr. \tag{S8}$$

The angular quadrature uses 16 Gauss–Legendre polar nodes and 32 azimuthal nodes, and the radial quadrature uses 32 Gauss–Legendre nodes on each of $[0, a]$ and $[a, b]$. The harmonics are defined in the Cartesian frame, with z normal to the layer. Because the angular projection retains the Bloch phase, it introduces an atom-dependent overall phase, which cancels in the density matrix. The spin components are rotated with the common rotation of Sec. S3.1. The projections include all channels through $\ell = 4$, and the total within the window is integrated separately.

The grouped d contributions sum the spin-density matrices of d_{z^2}, (d_{xz}, d_{yz}), and $(d_{xy}, d_{x^2-y^2})$, since grouping the two pairs reduces the dependence on the choice of in-plane orbital axes. Like the individual channels, the grouped quantities are angular projections of the spin density within the window. Tables S9 and S10 list the orbital fractions at each site and the changes relative to the corresponding pristine sites.

The coupling of these channels by SOC can be described in a local $L \cdot S$ representation, in which $L \cdot S = L_z S_z + (L_+ S_- + L_- S_+)/2$ and the transverse terms permit coupling between the $m = 0$ d_{z^2} harmonic and the $|m| = 1$ (d_{xz}, d_{yz}) subspace. We do not compute the corresponding matrix elements, and diagonal orbital projections do not contain the interorbital coherences needed to identify such a path, or to assign a specific Fe–Te hybridization mechanism from coincident PDOS features.

S4 SOC torque decomposition and numerical checks

To identify which atoms and angular-momentum channels exert the SOC torque on the spins near a vacancy, this section decomposes that torque by source atom and projector angular momentum (Fig. S6).

Vacancy	Fe plane	(x, y) (Å)	μ_r	d_{z^2} (%)	$d_{xz} + d_{yz}$ (%)	$d_{xy} + d_{x^2-y^2}$ (%)
Fe(1) vac.	central	$(+2.01, -1.16)$	-0.015842	56.30	15.92	26.89
Fe(1) vac.	central	$(0.00, +2.32)$	-0.019018	50.19	20.16	28.79
Fe(1) vac.	central	$(-2.01, -1.16)$	-0.015809	56.37	15.86	26.87
Fe(2) vac.	upper	$(-2.01, +1.16)$	-0.014905	18.89	30.03	50.81
Fe(2) vac.	upper	$(0.00, -2.32)$	-0.013353	20.09	29.85	49.91
Fe(2) vac.	upper	$(+2.01, +1.16)$	-0.015108	18.88	30.17	50.68
Fe(2) vac.	lower	$(-2.01, +1.16)$	0.014903	18.93	29.99	50.81
Fe(2) vac.	lower	$(0.00, -2.32)$	0.013250	20.16	29.71	50.01
Fe(2) vac.	lower	$(+2.01, +1.16)$	0.014696	18.93	29.83	50.98

Table S9: Occupied radial moments and orbital fractions at the nearest Fe sites used in the orbital comparison, all 2.62 Å from the removed atom. (x, y) is the in-plane position relative to the vacancy. μ_r is the radial component of the total spin moment within the window, in μ_B. Percentages divide the radial component of each orbital group by the total of that site. The remaining s, p, higher-ℓ, and unresolved angular weight is not assigned to a d channel. Both planes of the Fe(2)-vacancy neighborhood have the same leading orbital group, with opposite radial signs.

Vacancy	Fe plane	Sites	$\Delta\mu_r$	$\Delta\mu_z$	Leading radial (%)	Leading longitudinal (%)
Fe(1) vac.	central	2	-0.017419	-0.252384	52.99	39.33
Fe(2) vac.	upper	3	-0.014480	-0.298804	50.47	66.27
Fe(2) vac.	lower	3	0.014275	-0.298412	50.62	66.28

Table S10: Changes from the orbital projections at the corresponding pristine sites, averaged over the sites with pristine projections. Moment changes are in μ_B. The leading group is d_{z^2} for the central-plane Fe sites of the Fe(1) vacancy and $d_{xy} + d_{x^2-y^2}$ for the upper- and lower-plane Fe sites of the Fe(2) vacancy. The final columns divide the radial or longitudinal change of that group by the corresponding change of the total within the window. For the Fe(1) vacancy, two of the three corresponding pristine sites have orbital projections. The site LDOS averages in Fig. S2 include all three sites.

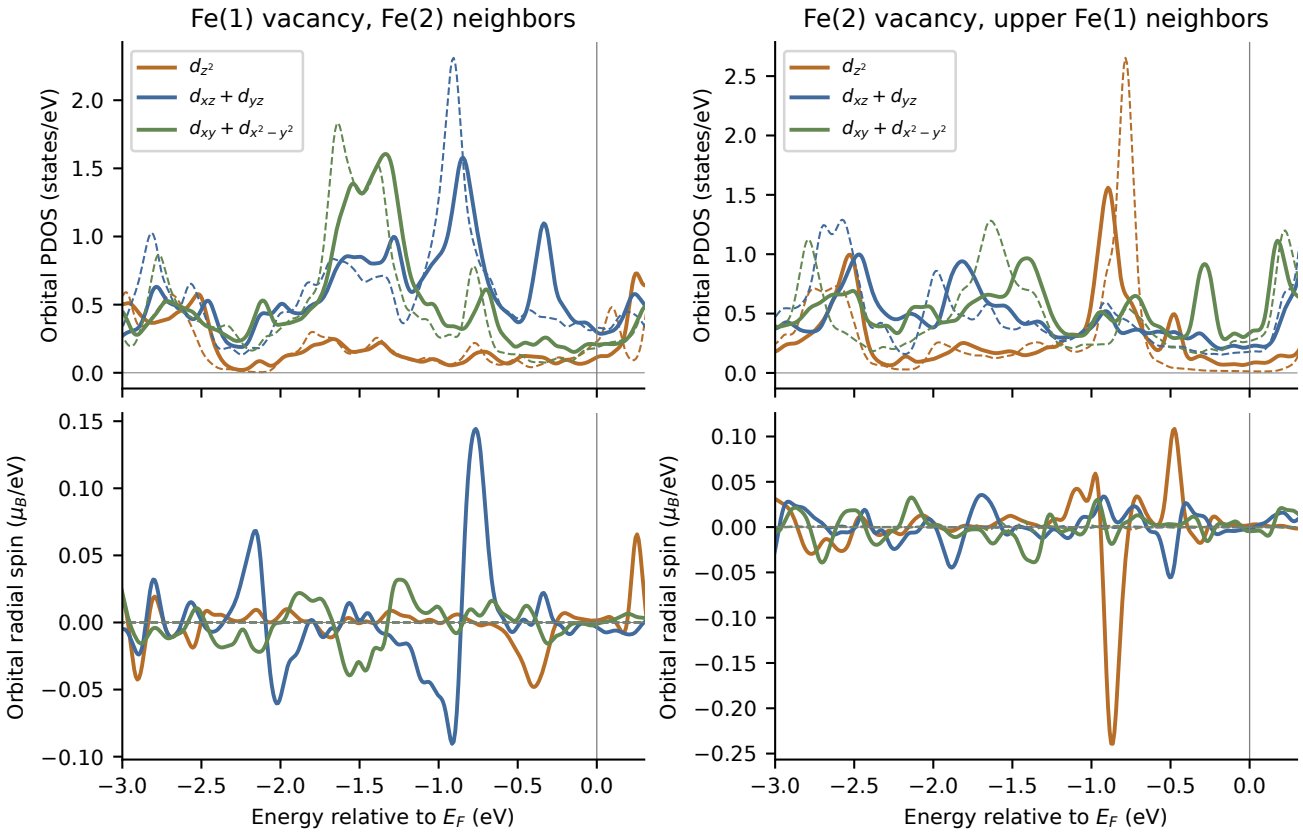


Figure S4: Orbital PDOS (top) and radial spin spectral density (bottom) at the largest-canting sites, each compared with its corresponding pristine site. The left panel shows the central-plane Fe(2) site beside the Fe(1) vacancy and the right panel the upper-plane Fe(1) site beside the Fe(2) vacancy. Solid and dashed curves are vacancy and pristine. Orbital frames, window, and $w = 0.05$-eV Gaussian smoothing are common to both panels.

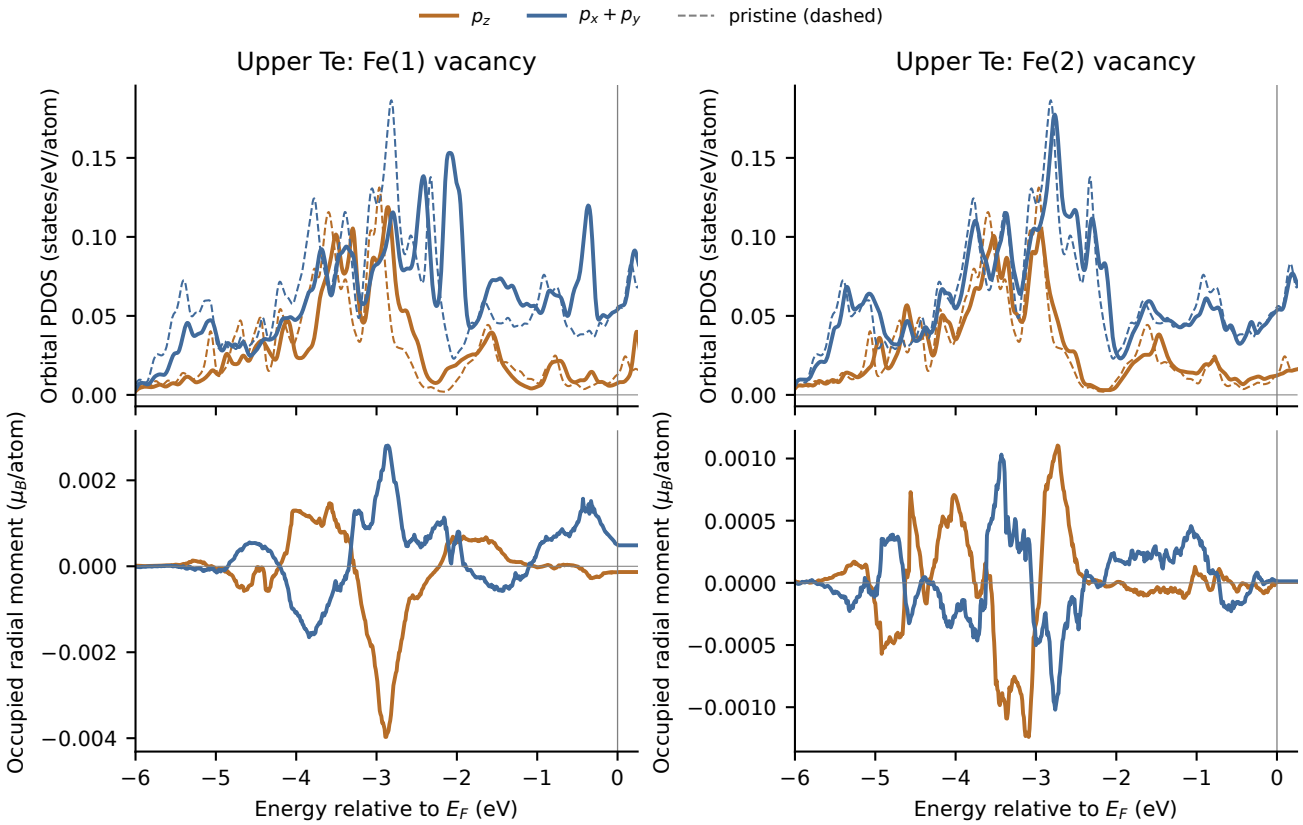


Figure S5: Upper-plane Te orbital PDOS (top) and cumulative occupied radial spin components (bottom), averaged over the three off-axis upper-plane Te atoms closest to the vacancy among those with orbital projections. Solid and dashed PDOS curves are vacancy and pristine. All cumulative curves are vacancy values and include all states below −6 eV. For the Fe(2) vacancy, the three Te atoms are three of the six off-axis Te atoms within 5 Å of the vacancy.

The torques are evaluated in the canted self-consistent states, whereas Eq. (1) of the main text involves energy derivatives at the collinear reference.

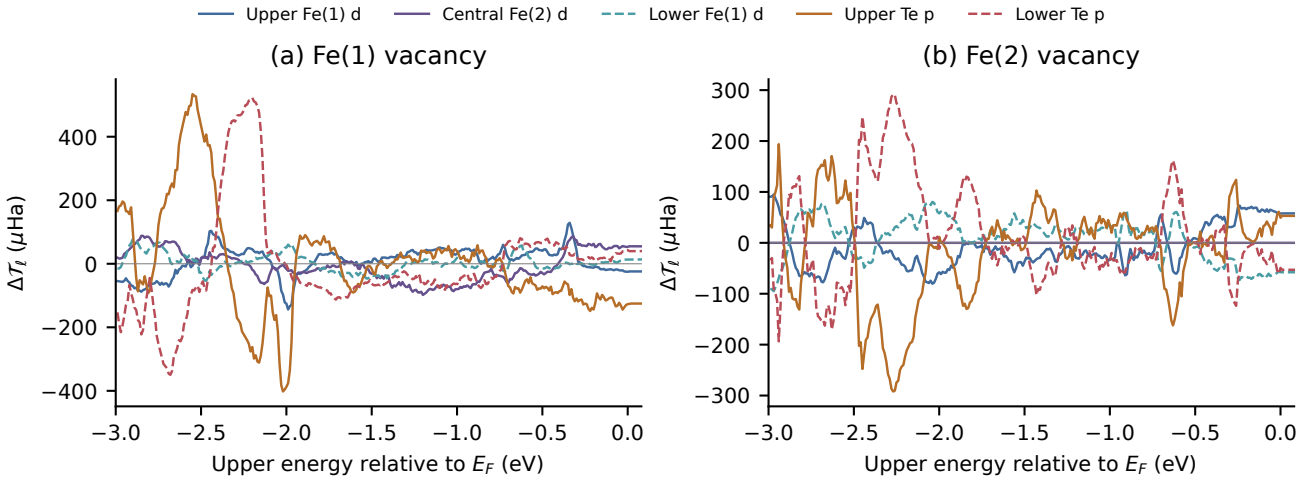


Figure S6: Cumulative SOC torque contributions $\Delta\mathcal{T}_\ell$, the occupation-weighted signed tangential torque of the source atoms in one plane and projector angular momentum ℓ, summed over states up to the energy on the horizontal axis, minus the same quantity for the corresponding pristine atoms.

We decompose the nonlocal pseudopotential into components $V_{I\ell}$ by atom I and projector angular momentum ℓ. For a Kohn–Sham state $n\mathbf{k}$, the SOC torque from component $I\ell$ is

$$\boldsymbol{\tau}_{n\mathbf{k},I\ell} = \mathrm{Re}\langle u_{n\mathbf{k}}|i[V_{I\ell},\mathbf{S}]|u_{n\mathbf{k}}\rangle, \qquad \mathbf{S} = \boldsymbol{\sigma}/2. \tag{S9}$$

Because the expectation value is taken over the full state, the index I labels the atom whose pseudopotential exerts the torque, without restricting the integral to a region around that atom. The spin-independent part of $V_{I\ell}$ commutes with $\mathbf{S}$. Here ℓ is the angular momentum of the nonlocal projectors, which is distinct from the orbital-projection channels of Sec. S3.4 and is not an adjustable SOC parameter.

For the Fe(2) vacancy, we sum over the atoms whose corresponding pristine positions lie within 5 Å of the vacancy in three dimensions and off its axis ($r_{xy} > 0.05$ Å), a set that contains six Fe and six Te atoms in each of the upper and lower planes. For the Fe(1) vacancy, the same criterion selects six Fe atoms in each of the three Fe planes, six Te atoms in the upper plane, and three in the lower plane. Before the sum over atoms, the torque on each atom is projected onto its tangential unit vector $\mathbf{e}_{\phi,I} = (-e_{r,y}, e_{r,x}, 0)$, so that the result measures a signed spatial pattern that can remain finite when the net Cartesian torque nearly cancels. The cumulative sums are weighted by $w_{\mathbf{k}} f_{n\mathbf{k}}$ with the 50-K occupations. Figure S6 displays the interval $[-3, 0.1]$ eV, with contributions below -3 eV included as a constant offset, and each curve is the vacancy result minus that of the corresponding pristine atoms. Table S11 lists the totals.

Without pristine subtraction, the cumulative upper-Te $\ell = 1$ torque within the displayed interval reaches about -292 μHa when integrated up to -2.27 eV, 5.52 times the magnitude of its final value of $+52.93$ μHa. The torque thus results from cancellation across the occupied energies, and we do not assign it to a single feature near the Fermi level. As a check, the occupation-weighted energy sums reproduce the separately computed totals for each I, ℓ and for each atom to numerical precision.

S4.1 Spin continuity in an atomic window

Because the time derivative of the spin within an atomic window vanishes for stationary states, the resulting window-resolved balance provides a numerical check of the computed Kohn–Sham states and

Source	Sites	Upper	Lower	Upper pristine
Fe(1), $\ell = 2$	6	57.8817	-57.8876	0.0010
Te, $\ell = 1$	6	52.9326	-52.9239	-0.0168

Table S11: Occupation-weighted signed tangential SOC torques on the atoms within 5 Å of the Fe(2) vacancy, in μHa ($\hbar = 1$). The upper and lower columns are vacancy values without pristine subtraction, and the last column is the upper-plane value at the corresponding pristine sites. "Sites" is the number of atoms per plane.

the implemented operators. We use it as such a check and not as a mechanism for canting. All expectation values are occupation-weighted sums over the occupied states, and for the spin in a real window W_A, stationarity gives

$$\frac{d\mathbf{S}_A}{dt} = \langle i[H, W_A\mathbf{S}]\rangle = 0. \tag{S10}$$

For a Hamiltonian with a nonlocal pseudopotential, the commutator separates into three terms,

$$\boldsymbol{\Phi}_A = \langle i[T, W_A]\mathbf{S}\rangle, \tag{S11}$$
$$\boldsymbol{\tau}_{\mathrm{SOC},A} = \tfrac{1}{2}\langle\{W_A, i[V_{\mathrm{NL}}, \mathbf{S}]\}\rangle, \tag{S12}$$
$$\mathbf{Q}_{\mathrm{NL},A} = \tfrac{1}{2}\langle\{i[V_{\mathrm{NL}}, W_A], \mathbf{S}\}\rangle. \tag{S13}$$

The first term is the kinetic spin-current flux through the tapered window, $\int \nabla W_A \cdot \mathbf{J}_S$, with inward flux taken as positive, the second is the SOC torque within the window, and the third arises from the spatial nonlocality of the pseudopotential. Because the spin-independent part of a nonlocal pseudopotential can contribute to $\mathbf{Q}_{\mathrm{NL},A}$, the third term is not exclusively a SOC effect.

For a nonlocal potential with kernel $V_{\mathrm{NL}}(\mathbf{r}, \mathbf{r}')$,

$$[V_{\mathrm{NL}}, W_A]\psi(\mathbf{r}) = \int d\mathbf{r}'\, V_{\mathrm{NL}}(\mathbf{r}, \mathbf{r}')[W_A(\mathbf{r}') - W_A(\mathbf{r})]\psi(\mathbf{r}'). \tag{S14}$$

The commutator thus couples amplitudes at points with different window weights. It vanishes for a local potential and for $W_A = 1$ over the whole cell, and its value depends on the choice of window. We refer to $\mathbf{Q}_{\mathrm{NL},A}$ as the nonlocal-pseudopotential term of the window-resolved balance. The general role of spin continuity in stationary SOC calculations is discussed in Ref. [19].

S4.2 Vanishing XC torque in the locally collinear approximation

In the locally collinear approximation used here, the XC field is parallel to the local magnetization of the input density [4, 20], so the local XC torque vanishes [21, 22]. At exact density self-consistency,

$$\mathbf{B}_{\mathrm{xc}}(\mathbf{r}) \parallel \mathbf{m}(\mathbf{r}), \qquad \boldsymbol{\tau}_{\mathrm{xc},A} = \int W_A\, \mathbf{B}_{\mathrm{xc}} \times \mathbf{m}\, d\mathbf{r} = 0. \tag{S15}$$

The result holds pointwise, also where the magnetization direction varies in space, and for this XC approximation the self-consistent balance is therefore

$$\boldsymbol{\Phi}_A + \boldsymbol{\tau}_{\mathrm{SOC},A} + \mathbf{Q}_{\mathrm{NL},A} = 0. \tag{S16}$$

A vanishing local XC torque is nevertheless compatible with nonzero intersite exchange interactions.

The XC torque that we evaluate numerically, by contrast, combines the XC field of the Hamiltonian, built from the input density, with the magnetization of the computed states,

$$\mathbf{D}_A = \int W_A\, \mathbf{B}_{\mathrm{xc}}[n_{\mathrm{in}}, \mathbf{m}_{\mathrm{in}}] \times \mathbf{m}_{\mathrm{states}}\, d\mathbf{r}. \tag{S17}$$

The two magnetizations can differ at finite SCF tolerance and after the NSCF refinement of the eigenvectors, and we use $\mathbf{D}_A$ as a consistency check between the field and the spin density, not as an intrinsic XC torque of the locally collinear functional. For the fixed Hamiltonian and the computed states, the four terms satisfy

$$\mathbf{r}_A = \mathbf{\Phi}_A + \boldsymbol{\tau}_{\mathrm{SOC},A} + \mathbf{Q}_{\mathrm{NL},A} + \mathbf{D}_A \simeq 0. \tag{S18}$$

The residual $\mathbf{r}_A$ can be small for eigenstates of a fixed Hamiltonian even before density self-consistency is reached. A small residual therefore tests the eigenstates and the implemented operators, but not SCF convergence, magnetic stability, exchange parameters, or canting energies.

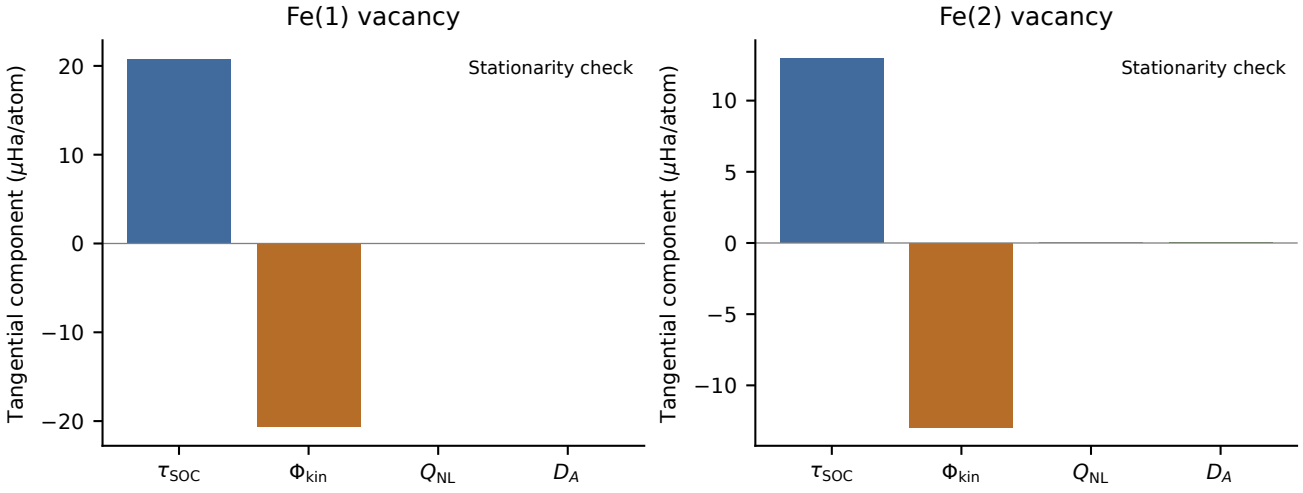


Figure S7: Window-resolved spin balance for the three-site Fe groups of Sec. S3.1. $\mathbf{D}_A$ is the cross product of the input XC field and the spin density of the computed states, Eq. (S17).

Vacancy	τ_{SOC}	Φ_{kin}	Q_{NL}	D_A	Max. residual norm
Fe(1) vac.	20.804098	-20.705350	0.000103	-0.097320	0.002389
Fe(2) vac.	12.940620	-12.950285	0.000211	0.008428	0.004643

Table S12: Tangential components of the terms in Eq. (S18), averaged per Fe atom over the three-site groups, in μHa with $\hbar = 1$. The final column is the largest norm of $\mathbf{r}_A$ among the three sites.

Figure S7 and Table S12 show the balance for the two groups, whose largest residual norms are 2.39×10^{-9} and 4.64×10^{-9} Ha for the Fe(1)- and Fe(2)-vacancy groups, respectively. The nonlocal-pseudopotential term $\mathbf{Q}_{\mathrm{NL},A}$ is small in these windows.

S5 Numerical convergence checks

This section collects the checks of the numerical accuracy of the canting angles, beginning with the refinement of the wave functions. Table S13 compares the Fe moments of the GS and NSCF calculations.

The largest change of an Fe moment direction between the GS and NSCF calculations is 0.001275°. The comparison tests the refinement of the wave functions for a converged SCF state, rather than an SCF calculation at tighter tolerance. The SCF convergence itself can be followed through the atomic moments at each iteration of the production calculations. Over the last five SCF iterations of the states magnetized along z, the direction of any Fe moment changes by at most 0.0013° in the pristine cell, 0.0043° for the Fe(1) vacancy, and 0.0011° for the Fe(2) vacancy, and the largest canting angles change

System	GS max.	NSCF max.	Max. vector change	Max. magnitude change
Pristine	0.014102	0.014073	0.000209	0.0000605
Fe(1) vacancy	0.948532	0.948481	0.000482	0.0000684
Fe(2) vacancy	0.444285	0.444107	0.001275	0.0001771

Table S13: GS and NSCF Fe moments with the cosine window. The first two columns are maximum canting angles. The maximum vector change is the largest angle between the GS and NSCF directions of the same Fe moment, rather than the difference of the largest canting angles. Angles are in degrees, and magnitude changes are in μ_B.

by 0.001° or less. Such changes indicate stable SCF convergence but do not bound the error of the reported states. The states magnetized along x are discussed in Sec. S6.

For the spatial discretization, our test calculations with seventh-order finite elements and a mesh size of 1.0 bohr near the atoms give maximum Fe angles of 0.01748°, 0.94555°, and 0.43837°, compared with the production values 0.01407°, 0.94848°, and 0.44411°. Because these test calculations also used intermediate geometries of the relaxation and different atomic windows, this comparison is not a controlled mesh-convergence study.

Within each production calculation, the canting angles of sites related by the local threefold axis of a vacancy differ by more than these mesh-related changes, with values of 0.790, 0.792, and 0.948° for the nearest Fe(2) of the Fe(1) vacancy and 0.385–0.444° for the nearest Fe(1) of the Fe(2) vacancy. Neither the rectangular cell nor the $4 \times 4 \times 1$ grid, whose reciprocal-space spacings differ along x and y, has this axis, and the cell vectors deviate from the ideal ratio by 0.09%. We take this spread, up to 20%, as the numerical uncertainty of the canting angles. In our test calculations (Table S14), performed at lower settings and at intermediate geometries of the relaxation, the maximum canting angle changes by 17% (Fe(2) vacancy) and 0.4% (Fe(1) vacancy, last iterate of an unconverged calculation) between the $2 \times 2 \times 1$ and $4 \times 4 \times 1$ grids, comparable to this spread. The symmetry constraints of Sec. S2.1, by contrast, are independent of the k-point grid and the cell. The spread is a diagnostic of the broken threefold symmetry, not a convergence bound. The two states compared in Sec. S6 use the same grid, cell, and geometry, but a change of the magnetization direction changes the bands and the response, so their numerical errors need not be correlated. The effect of the cell size on the canting far from the vacancy is discussed in Sec. S2.

System	2×2	4×4
Pristine	0.013083	0.007443
Fe(1) vacancy	1.119127	1.123167†
Fe(2) vacancy	0.453850	0.375260

Table S14: Maximum Fe canting angle, in degrees, from our test SCF calculations with different k-point grids. These calculations use sixth-order finite elements, pseudopotentials with 16, 14, and 16 valence electrons for Fe, Ge, and Te, and geometries that differ from the production structures. † marks the last iterate of an Fe(1)-vacancy calculation that did not converge. The table therefore characterizes the k-point dependence at lower settings, not a convergence series for the production settings.

S6 States magnetized along x

The main text separates the antisymmetric Dzyaloshinskii–Moriya interaction (DMI) and symmetric contributions to the canting around the vacancies by comparing the state magnetized along z with a state magnetized along x. This section describes these calculations and the relations on which the

comparison rests. We repeated the Fe(2)-vacancy SCF calculation with all Fe moments initialized along $+x$ at unchanged magnitude, using the relaxed geometry and the numerical settings of the state magnetized along z. The calculations for the states along x differ from those for the states along z in four respects. They used a later version of the code, 5600 instead of 5040 wave functions, FP32 instead of TF32 arithmetic in the Chebyshev filtering, and a final refinement of the wave functions within the SCF run instead of a separate NSCF calculation. The calculation converged in 60 SCF iterations to a density residual of 1.6×10^{-5}, below the tolerance of 2×10^{-5} used for the state magnetized along z. We compute the moments and SOC torques of this state, but not its vacuum maps or orbital projections. Its free energy exceeds that of the state magnetized along z by $E_x - E_z = 91.1$ meV per cell (89.8 meV in internal energy), or 0.85 meV per Fe atom.

With the magnetization along x, the mirror plane normal to x through the vacancy leaves m_x unchanged and reverses m_y and m_z. The summed moment therefore stays along x (computed tilt 0.0025°), and Fe atoms on this plane cannot cant. The horizontal mirror combined with time reversal leaves m_x and m_y unchanged and reverses m_z, so that the z components of the two outer planes are opposite and those of the central plane vanish.

To relate the two states, consider a bilinear model $E = \sum_{ij} \mathbf{e}_i \cdot \Gamma_{ij} \mathbf{e}_j$ with $\Gamma_{ji} = \Gamma_{ij}^T$ and single-site terms at $i = j$. The transverse field on atom i at a collinear reference $\hat{\mathbf{n}}$ is the component of $2\sum_j \Gamma_{ij}\hat{\mathbf{n}}$ perpendicular to $\hat{\mathbf{n}}$, whose x component for $\hat{\mathbf{n}} = \hat{\mathbf{z}}$ is $2\sum_j (\Gamma_{ij})_{xz}$ and whose z component for $\hat{\mathbf{n}} = \hat{\mathbf{x}}$ is $2\sum_j (\Gamma_{ij})_{zx}$. We take the linear response $\boldsymbol{\delta} = -\mathcal{K}^{-1}\mathbf{g}$ with the same stiffness $\mathcal{K}$ in the two states. The symmetric part of Γ then gives $\delta_{i,z}^{(x)} = \delta_{i,x}^{(z)}$, and the antisymmetric part gives $\delta_{i,z}^{(x)} = -\delta_{i,x}^{(z)}$, for every atom and including the response of neighbors to each other.

This comparison differs from the linear-response calculation of the exchange tensor. There, each magnetization direction gives the block of the pair tensor transverse to it, so that the xz and zx elements of a pair both follow from a reference along y [23, 24]. Here we use the first-order fields, which are sums over all partners of an atom, in the states along z and x. We have not validated the comparison against a calculation of the pair tensors.

These relations rest on three assumptions. First, the couplings are bilinear and do not depend on the magnetization direction. A higher-order single-site term such as $\kappa\, e_x e_z (e_z^2 - e_x^2)$, for example, gives $g_x^{(z)} = \kappa$ and $g_z^{(x)} = -\kappa$ and would therefore mimic the antisymmetric pattern without any intersite DMI. Chiral fourth-order interactions are allowed in this point group and have been reported for monolayer FGT [25, 26]. The comparison does not separate them from the DMI. Second, the compared components respond with the same stiffness. In the state magnetized along z, the response obeys $\mathcal{K}_{xx}^z \delta_x + \mathcal{K}_{xy}^z \delta_y = -g_x$, and in the state along x, $\mathcal{K}_{zz}^x \delta_z + \mathcal{K}_{zy}^x \delta_y = -g_z$, so the relations also require the SOC-induced mixing with the y components ($\mathcal{K}_{xy}$, $\mathcal{K}_{zy}$) to be negligible. The stiffness of the two states differs through SOC-derived terms of the order of the anisotropy energy, about 1 meV per Fe, but this comparison with the exchange does not bound the softest nonuniform mode. Third, the state magnetized along x is a hard-axis state. The mirror normal to x makes it stationary, and the SCF iterations did not rotate the magnetization away from it, so that only its response to the local fields, not its stability, enters the comparison. Writing the compared fields as $s \pm a$ and the two responses as $\bar{R} \pm \Delta R$, the measured combinations are $\delta^{\mathrm{a}} = \bar{R}a + \Delta R\, s$ and $\delta^{\mathrm{s}} = \bar{R}s + \Delta R\, a$, so deviations contaminate both. A large antisymmetric fraction therefore identifies a predominantly antisymmetric, DMI-type response of the xz element, not the coupling tensor itself. We write the antisymmetric part of each ordered pair as the DMI, $\mathbf{D}_{ij} \cdot (\mathbf{e}_i \times \mathbf{e}_j)$, counting each bond twice in the ordered sum, which gives $(\Gamma_{ij}^{\mathrm{a}})_{xz} = -(\mathbf{D}_{ij})_y$, and the comparison therefore tests only the y components of the DMI vectors. The y canting components in the two states involve different elements of Γ and are not compared.

Table S15 lists the Fe(1) atoms within 5 Å that lie off the mirror plane normal to x. For the four nearest neighbors, the ratio $\delta_z^{(x)}/\delta_x^{(z)}$ ranges from -0.78 to -0.81, and the correlation coefficient over these atoms is -0.9999. The antisymmetric fraction $\sum(\delta_i^{\mathrm{a}})^2 / \sum[(\delta_i^{\mathrm{a}})^2 + (\delta_i^{\mathrm{s}})^2]$ is 0.987 for the nearest

neighbors and 0.24 for the second Fe(1) shell. For the nearest neighbors, the fraction changes by less than 0.001 across the six integration windows of Table S1, and the symmetric part amounts to 11% of the antisymmetric part in amplitude. Their response is 19% to 22% weaker in the state along x. A uniaxial anisotropy makes the state along the hard axis softer against out-of-plane tilts, so that a difference in stiffness of this origin would give a stronger response. The weaker response therefore indicates a symmetric admixture of this size or couplings that depend on the magnetization direction. If the orbital moments are included in the atomic moments, the fraction for the nearest neighbors is 0.83, because the orbital moments of these atoms tilt in the same sense in both states, by 59 and 67 mrad. In the second shell, where the canting is at most 0.03°, $\delta_z^{(x)}$ is 3 to 5 times $\delta_x^{(z)}$, so the shell follows neither the antisymmetric pattern nor, quantitatively, the symmetric one. Its canting in the state along z is mostly tangential. The local threefold symmetry forbids this component for these atoms, so that it has the size of the numerical artifacts, and we do not classify this shell. On rotation of the magnetization from z to x, the SOC torque component τ_y keeps its sign on the nearest neighbors ($\pm$10.7 to $\pm$4.5 μHa) but reverses on the second shell ($\pm$4.6 to $\mp$3.1 μHa). Section S6.3 uses these torques quantitatively as a test that does not require equal stiffness.

Plane	x (Å)	y (Å)	$\delta_x^{(z)}$	$\delta_z^{(x)}$	δ^{s}	δ^{a}	$\tau_y^{(z)}$, $\tau_y^{(x)}$ (μHa)
lower	-2.01	+1.16	-6.97	+5.60	-0.68	-6.28	+10.74, +4.47
lower	+2.01	+1.16	+6.90	-5.61	+0.64	+6.26	-10.75, -4.47
upper	-2.01	+1.16	+7.04	-5.56	+0.74	+6.30	-10.74, -4.48
upper	+2.01	+1.16	-7.10	+5.56	-0.77	-6.33	+10.73, +4.48
lower	-4.01	-2.32	+0.37	+1.22	+0.79	-0.42	+4.59, -3.10
lower	+4.01	-2.32	-0.45	-1.22	-0.83	+0.39	-4.60, +3.11
upper	-4.01	-2.32	-0.33	-1.21	-0.77	+0.44	-4.60, +3.12
upper	+4.01	-2.32	+0.24	+1.21	+0.73	-0.48	+4.60, -3.11

Table S15: Canting components (mrad) of the off-mirror Fe(1) atoms within 5 Å of the Fe(2) vacancy in the states magnetized along z and x, with $\delta^{\mathrm{s}} = (\delta_x^{(z)} + \delta_z^{(x)})/2$ and $\delta^{\mathrm{a}} = (\delta_x^{(z)} - \delta_z^{(x)})/2$, and the y component of the SOC torque on each atom in the two states. Positions are relative to the vacancy.

In the state magnetized along x, the largest canting angle is 1.06°, on Fe(1) atoms 6.25 Å from the vacancy, mainly along y. Two of the six central-plane Fe(2) neighbors at 4.0 Å cant by 0.61° along y and the other four by 0.16–0.22°, which the symmetry of this state permits. These in-plane components were not yet stationary when the density residual reached the tolerance. Over the last twelve SCF iterations, the y component of the nearest Fe(1) neighbors grew from 7.3 to 8.2 mrad and the largest y component in the cell from 17.6 to 18.4 mrad, whereas the z components used in the comparison changed by less than 0.01 mrad. The in-plane angles given here are therefore lower bounds. Moment magnitudes differ between the two states by at most 0.037 μ_B. We use the state magnetized along x, which is not the ground state, only to separate the couplings.

S6.1 Fe(1) vacancy and pristine cell

The same comparison was carried out for the Fe(1) vacancy, and a pristine cell magnetized along x was computed as a reference for the anisotropy energy and the SOC torques. Both calculations use the relaxed geometries and production settings of the corresponding states magnetized along z, with the four differences listed above. They converged in 49 (Fe(1) vacancy) and 46 (pristine) SCF iterations to density residuals below 2×10^{-5}, and the summed Fe moments stay along x to within 0.0004° and 0.003°. The electronic free energies at 50 K give the anisotropy energy $E_x - E_z = 103.5$ meV per cell (0.958 meV per Fe) for the pristine cell, 91.1 meV (0.851 meV per Fe) for the Fe(2) vacancy, and 111.4 meV (1.041 meV per Fe) for the Fe(1) vacancy. The Fe(2) vacancy therefore lowers the anisotropy

energy of the cell by 12.4 meV, whereas the Fe(1) vacancy raises it by 7.9 meV. The internal energies give 100.5, 89.8, and 110.4 meV and vacancy-induced changes of −10.7 and +9.9 meV, so the difference between free and internal energy is not negligible on this scale. The states along z use TF32 and those along x FP32 arithmetic in the Chebyshev filtering. Replacing the free energies of the states along z by those of the FP32 NSCF recomputation of the same Hamiltonian changes them by +0.007 meV (pristine), +0.077 meV [Fe(2) vacancy], and −0.001 meV [Fe(1) vacancy], and the vacancy-induced changes by −0.070 and +0.008 meV. This check does not test a fully self-consistent density obtained with different arithmetic, but it shows no sensitivity that would alter the 8–12 meV effects.

The pristine value of 0.96 meV per Fe lies within the range of published values for the monolayer, which are 0.92 meV per Fe in the local-density approximation (LDA) [27], 1.3 meV per Fe [28], and 1.22 meV per Fe in the generalized-gradient approximation [29]. Vacancy-induced changes have been published for a cell of nine formula units with one vacancy [30]. With PBE they are +0.46 meV for the Fe(2) vacancy and +1.21 meV for the Fe(1) vacancy, and with the LDA they are −20.8 and −11.2 meV. Our value for the Fe(2) vacancy therefore differs in sign from the published PBE value. The two calculations differ in the cell size, the k-point sampling, the electronic temperature, the dispersion correction, and the pseudopotentials. The vacancy-induced changes are differences between four total energies and amount to about 0.1 meV per Fe. At the electronic temperature of 50 K, only about 70 states of the $4 \times 4 \times 1$ grid are partially occupied. We have not repeated the calculations with a denser grid or a higher electronic temperature. The vacancy-induced changes, including the sign for the Fe(2) vacancy, are therefore not yet established.

The Fe(1) vacancy lacks σ_h, so in the state magnetized along x the central plane may cant out of plane, and its Fe(2) neighbors enter the comparison. The mirror plane normal to x through the vacancy remains a symmetry, and atoms on it are again excluded. Table S16 lists the off-mirror Fe atoms within 5 Å. For the two nearest Fe(2) neighbors off the mirror plane, the ratio $\delta_z^{(x)}/\delta_x^{(z)}$ is −0.87, and the antisymmetric fraction is 0.996. It changes by less than 0.001 across the six integration windows and is 0.97 if the orbital moments are included. Over the last twelve SCF iterations of the state along x, the z component of these neighbors changed by 1%. The fraction is 0.66 for the same-plane Fe(1) atoms at 4.0 Å, 0.94 for the lower-plane Fe(1) atoms at 4.7 Å, and 0.88–0.99 in each distance range beyond 5.5 Å (Table S19). In the state magnetized along x, the largest canting angle is 1.73°.

S6.2 Radial and tangential components

The comparison gives only the x component of δ^{a} and δ^{s} for each atom. Within a shell, atoms related by the local threefold axis of the vacancy form orbits of three, on which a vector field is fixed by one radial and one tangential amplitude, $\delta_x = \delta_{\mathrm{rad}} \cos\phi - \delta_{\mathrm{tan}} \sin\phi$, where ϕ is the azimuth around the vacancy. We fit these two amplitudes for each orbit. Mirror-partner orbits have equal radial and opposite tangential amplitudes, as $\sigma_v T$ requires, and Table S17 lists one orbit of each mirror pair and every orbit that is its own mirror image. Around the Fe(2) vacancy, the antisymmetric canting of the nearest neighbors is purely radial and opposite in the two outer planes, as σ_h requires. Around the Fe(1) vacancy, the antisymmetric canting is radial and inward on the nearest central-plane neighbors and outward on the Fe(1) neighbors of both outer planes, and none of this radial antisymmetric canting has a mirror partner. This absence of a partner is what allows a net Néel-type DMI. It does not determine the coefficient of that DMI, which is a spatially weighted sum over bonds, or equivalently the slope of the spin-spiral energy at small wave vector, rather than a sum of the local fields obtained here. The same-plane Fe(1) atoms at 4.0 Å have tangential canting of comparable antisymmetric (4.5 mrad) and symmetric (3.9 mrad) amplitude. Because a mirror plane bisects the bond from these atoms to the vacant site, the removed bond produces only a radial field on them (main text), so their antisymmetric tangential canting comes from changes in the remaining bonds or from the response to fields on other atoms. For the lower-plane Fe(1) atoms at 4.7 Å, a twofold axis through the midpoint of the bond to

Plane	x (Å)	y (Å)	$\delta_x^{(z)}$	$\delta_z^{(x)}$	δ^{s}	δ^{a}	$\tau_y^{(z)}$, $\tau_y^{(x)}$ (μHa)
central	-2.01	-1.16	+12.46	-10.91	+0.78	+11.69	-17.61, -9.69
central	+2.01	-1.16	-12.47	+10.90	-0.78	-11.69	+17.60, +9.68
upper	-4.01	+0.00	-2.52	+1.16	-0.68	-1.84	+3.86, +5.15
upper	-2.01	-3.48	+3.94	+1.42	+2.68	+1.26	+4.21, +3.65
upper	-2.01	+3.48	-10.75	+2.47	-4.14	-6.61	-0.60, +0.48
upper	+2.01	-3.48	-3.88	-1.45	-2.67	-1.22	-4.23, -3.65
upper	+2.01	+3.48	+10.70	-2.50	+4.10	+6.60	+0.59, -0.48
upper	+4.01	+0.00	+2.47	-1.13	+0.67	+1.80	-3.86, -5.17
lower	-4.01	+0.00	-7.78	+5.08	-1.35	-6.43	-2.61, -0.53
lower	-2.01	-3.48	+4.09	-2.36	+0.86	+3.22	-0.61, -0.40
lower	-2.01	+3.48	-15.70	+9.40	-3.15	-12.55	-1.66, -0.45
lower	+2.01	-3.48	-4.06	+2.35	-0.86	-3.21	+0.61, +0.40
lower	+2.01	+3.48	+15.65	-9.41	+3.12	+12.53	+1.64, +0.46
lower	+4.01	+0.00	+7.75	-5.04	+1.35	+6.39	+2.60, +0.53
central	-4.01	+2.32	-5.53	+1.98	-1.77	-3.75	+2.43, -1.23
central	+4.01	+2.32	+5.46	-1.97	+1.75	+3.72	-2.44, +1.23

Table S16: Canting components (mrad) of the off-mirror Fe atoms within 5 Å of the Fe(1) vacancy in the states magnetized along z and x, with δ^{s} and δ^{a} as in Table S15, and the y component of the SOC torque that each atom exerts on its own spin.

the vacant site, perpendicular to the bond, makes $\mathbf{D}_{i0}$ perpendicular to that axis, so that Eq. (2) of the main text gives a purely tangential field. These atoms indeed have a large antisymmetric tangential amplitude, but also a comparable radial one, while the DMI-type SOC torque on their own spin is small (RMS 1.1 μHa, against 13.7 μHa on the nearest Fe(2) neighbors). Their canting is therefore consistent with the exchange-mediated response to fields on other atoms, although the own-site torque omits fields from the SOC of neighboring Te atoms (Sec. S6.3). The fits assume the local threefold symmetry, which the rectangular cell breaks at the level of the 20% spread in Sec. S5, and the three-atom orbits at 4.0 and 4.7 Å leave residuals of about 1 mrad per atom (Table S17).

S6.3 Source-resolved SOC torques and spatial extent

The SOC torque exerted by the pseudopotential of each atom on the total spin, evaluated in both states, separates in the same way into a DMI-type part, which keeps its sign when the magnetization is rotated, and a symmetric part, which reverses. Subtracting the torques at the matched sites of the pristine cell changes these values by at most 0.03 μHa. This holds for the y component, which the mirror normal to x through every pristine atom forbids. The x component is not protected by an exact symmetry of the rectangular cell. In the pristine state magnetized along z it is uniform within each plane, with -3.8 and $+3.8$ μHa on the upper and lower Te atoms, and reflects the broken threefold symmetry of the cell and the k-point grid. We do not use it. Table S18 gives the root-mean-square parts for the off-mirror atoms within 5.2 Å. Around both vacancies, the Fe torques are mostly DMI-type (70–98%), whereas the Te torques are mostly symmetric (2–20%). In magnitude, however, the DMI-type Te torques are comparable to the Fe ones, with summed squares of 365.9 against 233.6 $\mu\mathrm{Ha}^2$ for the Fe(2) vacancy and 469.4 against 452.3 $\mu\mathrm{Ha}^2$ for the Fe(1) vacancy. The SOC of both species therefore contributes to the DMI-type torque pattern, and the Te SOC also produces the large symmetric torques. The Te torques act on the small induced spin near the Te atoms and reach the Fe moments only through hopping. The decomposition refers to the self-consistent states, in which the SOC of all species acts together, and it is not equivalent to removing the SOC of one species.

Vacancy	Plane	r (Å)	$\delta^{\mathrm{a}}_{\mathrm{rad}}$	$\delta^{\mathrm{a}}_{\mathrm{tan}}$	$\delta^{\mathrm{s}}_{\mathrm{rad}}$	$\delta^{\mathrm{s}}_{\mathrm{tan}}$	residual
Fe(2)	lower	2.62	+7.2	0.0	+0.8	0.0	0.0
Fe(2)	upper	2.62	-7.3	0.0	-0.9	0.0	0.0
Fe(2)	upper	4.79	-0.5	0.0	+0.9	0.0	0.0
Fe(2)	lower	4.79	+0.5	0.0	-0.9	0.0	0.0
Fe(2)	upper	6.25	-1.2	2.4	+0.7	0.1	0.2
Fe(2)	lower	6.25	+1.1	2.5	-0.8	0.1	0.1
Fe(1)	central	2.62	-13.5	0.0	-0.9	0.0	0.0
Fe(1)	upper	4.01	+3.0	4.5	+0.9	3.9	1.2
Fe(1)	lower	4.70	+7.4	9.1	+1.7	2.3	1.0
Fe(1)	central	4.79	+4.3	0.0	+2.0	0.0	0.0
Fe(1)	central	6.25	+6.2	6.1	-0.0	0.7	0.3
Fe(1)	upper	6.95	+4.6	0.0	+0.1	0.0	0.0
Fe(1)	upper	6.95	+3.0	0.0	-0.7	0.0	0.0

Table S17: Radial and tangential amplitudes (mrad) of the antisymmetric (δ^{a}) and symmetric (δ^{s}) canting for one orbit per shell. Positive radial amplitudes point away from the vacancy. Tangential amplitudes are given in magnitude because they change sign between mirror-partner orbits, and only one orbit of each mirror pair is listed. For orbits that contain an atom on the mirror plane normal to x, the tangential amplitude vanishes by symmetry and the fit uses the two remaining atoms. The last column is the RMS residual of the antisymmetric fit, which vanishes by construction for two-point fits.

Vacancy	Source	n	DMI-type	Symmetric	DMI fraction
Fe(2)	Fe(1), each outer plane	4	5.40	3.51	0.70
Fe(2)	Te, each outer plane	6	5.52	21.0	0.06
Fe(1)	Fe(2), central plane	4	9.66	3.08	0.91
Fe(1)	Fe(1), upper plane	6	3.46	0.52	0.98
Fe(1)	Fe(1), lower plane	6	1.13	0.69	0.73
Fe(1)	Ge	4	1.17	0.66	0.76
Fe(1)	Te, upper plane	4	10.79	21.51	0.20
Fe(1)	Te, lower plane	2	1.41	9.71	0.02

Table S18: Root-mean-square DMI-type and symmetric parts of the y component of the source-resolved SOC torques (μHa), pristine values subtracted, for the off-mirror atoms within 5.2 Å of each vacancy. For the Fe(2) vacancy, the central-plane Fe and Ge torques vanish by symmetry.

Test without the equal-stiffness assumption. The SOC torque that the pseudopotential of an Fe atom exerts on the spin in its own window gives the transverse field from that atom's SOC alone, without passing through the stiffness. It is a partial operator that omits the SOC of all other atoms, Fe, Ge, and Te, although its value depends on the self-consistent state in which all SOC acts. In a stationary state this torque is balanced by the spin flux into the window, which carries the exchange response (Sec. S4.1), so the canting does not enter its definition. Because it is evaluated in the canted state, it differs from the field at the collinear reference by terms of order the anisotropy stiffness times the canting angle. Using twice the cell-averaged anisotropy energy as a stiffness, $2K = 63$ and 77 μHa per radian for the Fe(2) and Fe(1) vacancies, and the largest canting angle of the nearest neighbors off the mirror plane in either state (0.57° and 0.79°), this correction is estimated at 0.6 and 1.1 μHa, compared with DMI-type torques of 7.6 and 13.7 μHa. For the same-plane Fe(1) atoms at 4.0 Å of the Fe(1) vacancy, which cant by up to 1.7° in the state along x, the estimate is 2.3 μHa against a DMI-type torque of 3.5 μHa. These estimates use a cell average and are not a bound on the local response. The same sign rule applies, so the DMI-type part keeps its sign when the magnetization is rotated from z to x, and the symmetric part reverses. The fields from the SOC of other atoms enter the window through the spin flux and are not contained in this torque. The canting split and the torque split therefore rest on different approximations, equal stiffness for the first and the omission of the SOC of all other atoms for the second. Both retain the bilinear interpretation, and the higher-order term above would also give a torque of the same sign in the two states, so the torque test corroborates the sign character of the nearest-neighbor response rather than excluding higher-order anisotropy. Where the local fields are largest, at the nearest neighbors of both vacancies, the two agree, with DMI-type fractions of 0.99 (canting) and 0.86 (torque) for the Fe(2) vacancy and 1.00 and 0.92 for the Fe(1) vacancy. Both are mostly symmetric in the second Fe(1) shell of the Fe(2) vacancy (0.24 and 0.04). Elsewhere the two fractions differ (Table S19). For the same-plane Fe(1) atoms at 4.0 Å of the Fe(1) vacancy, the torque is 98% DMI-type whereas the canting is 66% antisymmetric. There the canted-state correction is comparable to the field, and the symmetric tangential canting may also come from the SOC of the neighboring upper Te atoms, whose torques are 80% symmetric (Table S18) but do not enter the own-site torque. On shells where the own-site torque is small, such as the lower-plane Fe(1) atoms at 4.7 Å and the central-plane atoms at 4.8 Å, the canting is consistent with the response to fields on other atoms, and the two fractions need not agree.

Table S19 compares, shell by shell, the antisymmetric fraction of the canting with that of the SOC torque that each Fe atom exerts on its own spin. The DMI-type torque is large only on the nearest one or two shells (7.6 μHa at 2.62 Å around the Fe(2) vacancy, and 13.7 and 3.5 μHa at 2.62 and 4.0 Å around the Fe(1) vacancy) and at most about 1 μHa beyond. The antisymmetric canting, by contrast, persists to the edge of the cell, where the torques are small and of mixed character. The DMI-type torque from Fe SOC is therefore localized near the vacancy. The Te source torques are not. Over the whole cell, with the same selection and pristine subtraction, 74% (Fe(2) vacancy) and 38% (Fe(1) vacancy) of the summed squared DMI-type Te torque lies beyond 5.2 Å, with RMS values of 6.3 μHa at 7–9 Å around the Fe(2) vacancy. The corresponding fractions for the Fe torques are 14% and 5%. The complete SOC source pattern is therefore not confined to the nearest shells, and beyond about 6 Å it and the canting both include contributions from the periodic images.

S6.4 Estimate of the net DMI from the SOC torques

The comparison above identifies the DMI-type character of the local fields. It does not give the net DMI that a layer acquires when Fe(1) vacancies are distributed unequally between its outer planes. This subsection estimates that quantity from the SOC torques.

Consider a vacancy in a layer whose net DMI vanishes, and a cycloidal spin spiral with the moment directions $\mathbf{e}(\mathbf{r}) = (\sin qx, 0, \cos qx)$, where x is measured from the vacancy. The spiral rotates the moments about y by the angle $\varphi = qx$ and propagates along x. A rotation of the spin at atom I by

	Fe(2) vacancy			Fe(1) vacancy		
r (Å)	n	F_a (canting)	F_a (torque)	n	F_a (canting)	F_a (torque)
0–3	4	0.99	0.86	2	1.00	0.92
3–4.5	–	–	–	6	0.66	0.98
4.5–5.5	4	0.24	0.04	8	0.93	0.47
5.5–7	12	0.92	0.38	10	0.99	0.36
7–9	12	0.80	0.20	22	0.98	0.23
9–11	16	0.85	0.52	28	0.88	0.42
11–16	18	0.53	0.76	23	0.95	0.75

Table S19: Antisymmetric fraction $F_\mathrm{a} = \sum(\delta^\mathrm{a})^2 / \sum[(\delta^\mathrm{a})^2 + (\delta^\mathrm{s})^2]$ of the canting and the corresponding fraction of the SOC torque that each Fe atom exerts on its own spin, for the off-mirror Fe atoms in each distance range, with n the number of atoms. Central-plane atoms of the Fe(2) vacancy are excluded because they vanish by symmetry.

a small angle φ_I about y changes the SOC energy by $-\varphi_I \tau_{I,y}$, where $\boldsymbol{\tau}_I$ is the SOC torque exerted by the pseudopotential of atom I at the collinear reference (Sec. S4). To first order in q, the energy of the spiral relative to the collinear state is therefore

$$E^{(1)} = -q \sum_I x_I \, \tau_{I,y}. \tag{S19}$$

The sum is the first moment of the torques about the vacancy. It is well defined when the torques are confined to the neighborhood of the vacancy and the pristine layer contributes nothing. Along the spiral, the reference direction rotates from z to x. The DMI-type part of $\tau_{I,y}$ keeps its sign under this rotation, whereas the symmetric part reverses (Sec. S6.3) and averages to zero. Only the DMI-type part, $\tau^\mathrm{D}_{I,y} = (\tau^{(z)}_{I,y} + \tau^{(x)}_{I,y})/2$, therefore enters. For the DMI energy density $D(m_z \partial_x m_x - m_x \partial_x m_z)$, the spiral has the energy DqA in a cell of area A, so that one vacancy contributes

$$p = -\sum_I x_I \, \tau^\mathrm{D}_{I,y} \tag{S20}$$

to DA. In the dilute limit, areal densities n_u and n_l of Fe(1) vacancies in the upper and lower planes then give $D = p\,(n_\mathrm{u} - n_\mathrm{l})$. Equation (S20) is the atomic form of the relation between the DMI and the first moment of the torque [31]. To first order in SOC, this relation is equivalent to the one between the DMI and the ground-state spin current [32, 33].

We evaluate p from the source-resolved torques of all Fe, Ge, and Te atoms in the states magnetized along z and x, after subtracting the torques at the corresponding pristine sites. The positions x_I are the pristine positions relative to the vacancy, with minimum periodic images. Table S20 lists p as a function of the cutoff radius of the sum.

Three checks support the evaluation. For the Fe(2) vacancy, σ_h requires $p = 0$. The atoms of the upper and lower planes contribute -371.4 and $+370.6$ μHa Å, and their sum is 0.1% of their magnitudes. For the pristine cell, the mirror normal to x through every atom forbids $\tau_{I,y}$. The same sum about each of the 108 Fe sites of the pristine cell, without subtraction, has a root-mean-square value of 0.7 μHa Å and a largest magnitude of 1.4 μHa Å. For the ground state of the Fe(1) vacancy, the first moment of the full torques, which include the symmetric part, is -12 μHa Å. In this state, the symmetric torques that act on the canted moments thus nearly compensate the DMI-type torques in the sum, and the separation into the two parts is essential.

For the Fe(1) vacancy, p is negative for every cutoff radius, but it does not converge within the cell. It lies between -49 and -107 μHa Å (-1.3 to -2.9 meV Å) for cutoffs between 5 and 9.5 Å and reaches -224 μHa Å (-6.1 meV Å) for the whole cell. Three quarters of the whole-cell value come from atoms

Cutoff radius (Å)	Fe(2) vacancy	Fe(1) vacancy
3.0	0.0	−16.2
5.2	0.0	−67.6
6.5	+0.1	−107.0
8.0	−0.1	−50.7
9.5	−0.6	−49.3
11.0	−1.1	−141.8
12.5	−0.9	−232.6
14.0	−1.0	−216.2
whole cell	−0.8	−224.0

Table S20: First moment p of the DMI-type SOC torques about the vacancy, Eq. (S20), in μHa Å, summed over all atoms within the given three-dimensional distance of the vacancy. One μHa Å is 0.0272 meV Å.

within 2 Å of the cell boundary in x, for which the assignment of a periodic image is ambiguous, and the Te atoms contribute −199 μHa Å to it. We attribute this lack of convergence to the separate relaxation of the two states. Their canting differs away from the vacancy, and the in-plane canting of the states along x is larger (Sec. S6), so that the symmetric torques on the canted moments do not cancel in $\tau^{\mathrm{D}}_{I,y}$ there. A calculation for the reference along x in which the magnetization of the state along z is rotated rigidly would remove this difference. We have not performed it.

With one Fe(1) site per plane and formula unit, of area 13.95 Å^2, an excess of vacancies in the upper plane that amounts to a fraction c of its sites gives $D = pc/(13.95\ \text{Å}^2)$. For $c = 0.05$, the range of p from −1.3 to −6.1 meV Å corresponds to $|D| = 0.005$ to 0.022 meV/Å. Divided by the thickness of one layer, 8.2 Å, which is half the c lattice parameter of the bulk crystal [16], this is 0.01 to 0.04 mJ/m^2. The sign of D reverses for an excess in the lower plane.

The estimate has four limitations. It assigns each torque to the position of its atom and neglects the first moment of the torque within the atom. The relation between the first moment of the torque and the DMI is exact only to first order in SOC [33], and the SOC of Te is not weak. It uses two separately relaxed states, as discussed above. It refers to one vacancy per cell and to a spiral that propagates along x. The value for propagation along y follows from the threefold symmetry and would require a state magnetized along y to compute. We therefore regard the sign and the order of magnitude of p as the result. Its value varies by a factor of about four with the cutoff radius.

S7 Vacuum spin polarization

From the same self-consistent states, this section computes the spin polarization in the vacuum above the vacancies and evaluates it in a model for spin-polarized scanning tunneling microscopy (STM). We first summarize the result. For a tip polarization $P_t = 0.3$, the radial asymmetry difference [Eq. (S22)] of the Fe(1) vacancy is +2.16 to +3.64 percentage points over biases of +0.05 to +0.20 V and heights of 5–8 bohr, whereas for the Fe(2) vacancy it is +1.05 to +1.53 points at +0.10 V and −0.526 to −0.488 points at +0.20 V. Both ranges depend on the electronic-structure approximation and on the tip model described below.

The charge and spin spectral densities in the vacuum are computed directly from the spinors on a 64×64 Cartesian grid spanning the supercell in each plane parallel to the layer. For an energy window $\mathcal{W}$, let

$$N(\mathbf{r};\mathcal{W}) = \sum_{n\mathbf{k}} w_{\mathbf{k}} \chi_{\mathcal{W}}(\varepsilon_{n\mathbf{k}} - E_F) u^{\dagger}_{n\mathbf{k}}(\mathbf{r}) u_{n\mathbf{k}}(\mathbf{r}), \quad \mathbf{M}(\mathbf{r};\mathcal{W}) = \sum_{n\mathbf{k}} w_{\mathbf{k}} \chi_{\mathcal{W}}(\varepsilon_{n\mathbf{k}} - E_F) u^{\dagger}_{n\mathbf{k}}(\mathbf{r}) \boldsymbol{\sigma} u_{n\mathbf{k}}(\mathbf{r}). \quad \text{(S21)}$$

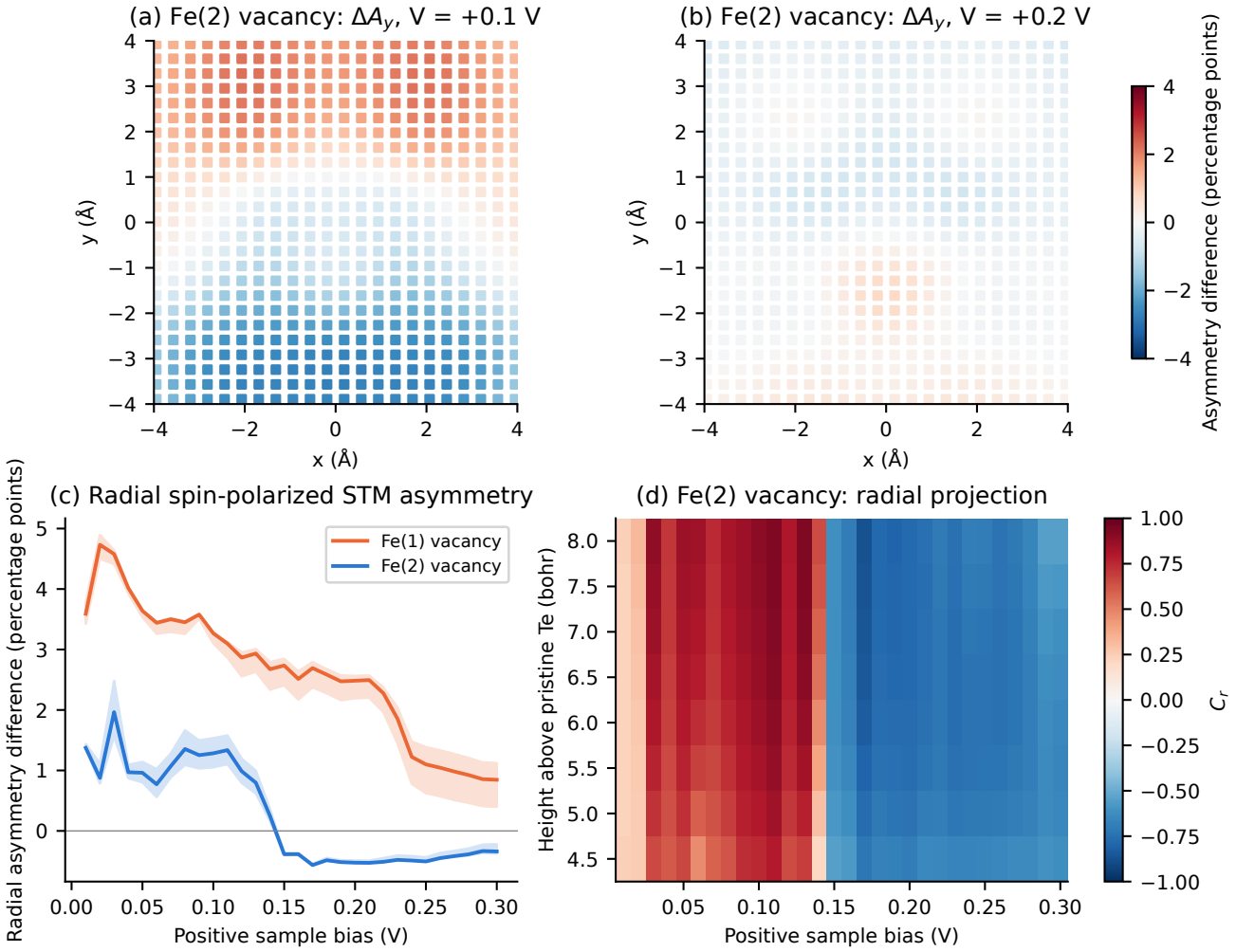


Figure S8: Vacuum spin polarization. (a,b) Fe(2)-vacancy minus pristine asymmetry for a tip polarized along y, in the spin-polarized Tersoff–Hamann model at 6 bohr above the pristine top-Te plane. (c) Radial asymmetry difference in the annulus, with shading spanning heights of 5–8 bohr. (d) Normalized radial projection C_r for the Fe(2) vacancy. The atomic moments are the same at all biases, and only the energy window changes.

Here $\mathbf{P} = \mathbf{M}/N$ is the window-integrated vacuum spin polarization. A positive sample bias corresponds to the unoccupied energies $\mathcal{W} = [0, eV)$ and a negative bias to the occupied energies $[eV, 0)$, with energies referred to the Fermi level of each system. The maps are computed for 100 adjacent 10-meV windows spanning $[-0.5, 0.5]$ eV, and each bias window is a sum of these bins, without Gaussian broadening or weighting by the GS occupations. The electronic temperature enters through the self-consistent state, and the effect of thermal broadening of the bias window is tested separately at the off-axis probes of Table S21.

For a spin-conserving s-wave tip with an energy-independent density of states and polarization $P_t\mathbf{a}$, the spin-polarized Tersoff–Hamann approximation [34–36] gives a current proportional to $N + P_t\mathbf{a} \cdot \mathbf{M}$ and hence an asymmetry $A_{\mathbf{a}} = P_t\mathbf{a} \cdot \mathbf{P}$. All results assume that tip reversal leaves the electronic state of the sample unchanged. We use $P_t = 0.3$ as a model parameter that sets the scale of the asymmetry, and the calculations specify neither an absolute current nor a tip material.

All heights are measured from the same pristine top-Te reference plane. At a given height, N and each component of $\mathbf{M}$ are interpolated linearly between the two nearest computed heights before $\mathbf{P}$ is formed, and vacancy and pristine maps are subtracted at identical Cartesian positions. The full scan (Fig. S9) covers two vacancies, 100 nonzero biases in 0.01-V steps, and eight heights from 4.5 to 8.0 bohr in 0.5-bohr steps, 1600 combinations in total. Within this scan, the Fe(1)-vacancy range quoted at the beginning of this section covers +0.05 to +0.20 V and heights from 5.0 to 8.0 bohr (2.65–4.23 Å above the reference plane).

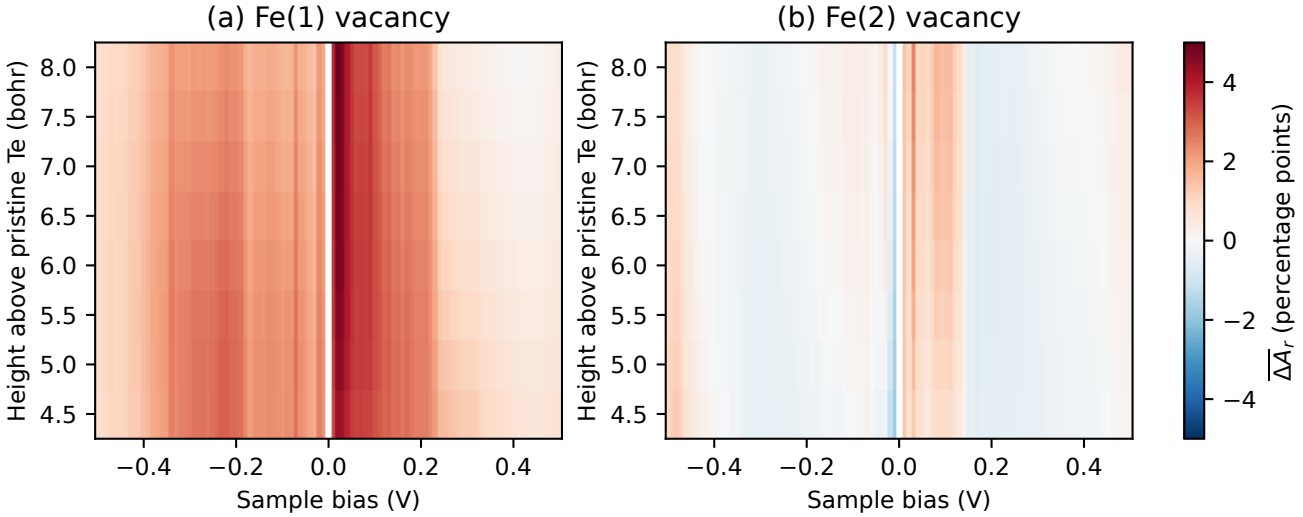


Figure S9: Bias and height dependence of the radial asymmetry difference in the fixed annulus for $P_t = 0.3$. Each column is one bias window and each row one height. The column at $V = 0$ is blank because its integration window is empty. The spatial masks and interpolation are the same throughout, and the positive-bias interval quoted in the text is a subset of this scan.

To condense each map into a few numbers, we average over the annulus $0.6 \le r_{xy} \le 3$ Å, which is fixed for each vacancy, with the same pixels at every bias and height. Let $w_j = N_{\mathrm{vac},j}$, $\mathbf{d}_j = \mathbf{P}_{\mathrm{vac},\perp,j} - \mathbf{P}_{0,\perp,j}$, and $d_{r,j} = \mathbf{d}_j \cdot \mathbf{e}_{r,j}$. The signed radial asymmetry difference and its normalized radial projection are then

$$\overline{\Delta A}_r = P_t \frac{\sum_j w_j d_{r,j}}{\sum_j w_j}, \qquad C_r = \frac{\sum_j w_j d_{r,j}}{\sqrt{(\sum_j w_j |\mathbf{d}_j|^2)(\sum_j w_j)}}, \tag{S22}$$

where C_r is the uncentered, normalized inner product of the polarization difference with the radial

unit-vector field. In addition, we compute

$$A_{\perp,\mathrm{rms}} = P_t \sqrt{\frac{\sum_j w_j |\mathbf{d}_j|^2}{\sum_j w_j}}, \qquad F_r = \frac{\sum_j w_j d_{r,j}^2}{\sum_j w_j |\mathbf{d}_j|^2}, \tag{S23}$$

and the exact relation $\overline{\Delta A}_r = C_r A_{\perp,\mathrm{rms}}$ provides a consistency check.

For each of 12 in-plane tip azimuths $\phi = 0°, 15°, \ldots, 165°$, the calculated map is $P_t(d_x \cos\phi + d_y \sin\phi)$, and its weighted normalized inner product with $\cos(\vartheta_j - \phi)$ measures the radial two-lobe pattern for that uniform tip direction. The absolute value of this inner product exceeds 0.875 throughout the Fe(1)-vacancy interval quoted above, and the radial fraction F_r ranges from 0.910 to 0.985. Both bounds apply to the tested heights, biases, tip model, and spatial mask. The radial quantities combine maps for several uniform tip directions into a vector pattern, rather than describing a tip with a spatially varying radial polarization. To test how this pattern changes with height, we compare, within the fixed annulus and at a single bias, the vector pattern at each integer height (5, 6, 7, and 8 bohr) with the 6-bohr pattern. For the biases +0.05, +0.10, +0.15, and +0.20 V, their unweighted inner product is at least 0.9918 for the Fe(1) vacancy and 0.9548 for the Fe(2) vacancy.

The processing of the maps was checked in two ways. A direct sum over the 10-meV energy bins, with triangular height-interpolation weights, reproduces ten selected entries of the scan at positive and negative biases to numerical precision, and all computed spectral density matrices satisfy $N \geq |\mathbf{M}|$ and $N \geq 0$. These checks test the internal consistency of the processing and do not estimate discretization errors.

We also test the sensitivity to k-point sampling, without new self-consistent calculations, at off-axis vacuum points 6 bohr above the reference plane. For the Fe(1) vacancy, the three probes lie above the upper-plane Te atoms of Fig. S5. For the Fe(2) vacancy, they lie above the three upper-plane Fe(1) neighbors at 2.62 Å of Sec. S3.1. The in-plane probe positions follow the relaxed atoms, which distinguishes these probes from the fixed annulus used for the maps, whereas the radial directions are defined by the pristine positions of the vacancy and of the probe sites. The density-weighted average over the three probes uses the formula for $\overline{\Delta A}_r$.

We use either all 16 k points, one of the four 2×2 subsets offset by parity, or the 15 points that remain after omitting one k point, with renormalized weights. Each selection is evaluated both with the self-consistent Fermi level and with a Fermi level refitted to preserve the full-grid electron count at 50 K. The bias windows are evaluated at temperatures $T_w = 0$, 50, and 120 K, where at finite T_w the sharp window is replaced by

$$|f_{T_w}(E - \mu - eV) - f_{T_w}(E - \mu)|. \tag{S24}$$

The chemical potential is always refitted at 50 K, independently of T_w, and none of these choices changes the density or the Hamiltonian.

For the Fe(2) vacancy, the radial asymmetry difference is positive at +0.10 V and negative at +0.20 V in every test (Table S21), whereas the signs at +0.05 and +0.15 V change across the 2×2 subsets. The tests therefore support the signs at +0.10 and +0.20 V but do not determine the voltage at which the sign changes. Because they use the fixed self-consistent states, they are not a substitute for a self-consistent convergence study of the k-point sampling.

S7.1 Unoccupied orbital comparison for the surface signal

To compare with the bias dependence of the vacuum polarization, we apply the orbital projections of Sec. S3.4 to the unoccupied states at positive energies.

For the Fe(2) vacancy, the comparison averages over the three upper-plane Fe(1) neighbors ($0.6 < r_{xy} < 3$ Å) and three upper-plane Te atoms with $0.6 < r_{xy} < 5$ Å, all within $r < 5$ Å of the vacancy. The Fe sites are those of Sec. S3.1, and the Te sites are those of Fig. S5. Because orbital projections

System	V (V)	Full grid	Leave one out	2×2 subsets
Fe(1) vacancy	+0.05	4.363, 4.649	4.193, 4.945	2.798, 6.066
Fe(1) vacancy	+0.10	3.844, 4.012	3.729, 4.373	3.289, 4.435
Fe(1) vacancy	+0.15	3.489, 3.672	3.420, 3.805	3.288, 3.958
Fe(1) vacancy	+0.20	3.086, 3.341	3.022, 3.452	2.952, 3.534
Fe(2) vacancy	+0.05	0.620, 0.930	0.291, 1.251	−0.746, 1.538
Fe(2) vacancy	+0.10	1.175, 1.492	0.986, 1.715	0.599, 2.844
Fe(2) vacancy	+0.15	−0.672, −0.379	−0.825, −0.264	−1.111, 0.422
Fe(2) vacancy	+0.20	−0.977, −0.906	−1.078, −0.835	−1.466, −0.725

Table S21: Ranges of the radial asymmetry difference at the off-axis probes, in percentage points for $P_t = 0.3$. Each range combines both Fermi-level conventions and all three window temperatures. Per bias, the full-grid column contains 6 values, the leave-one-out column 96 values, and the column of the four 2×2 subsets 24 values.

are computed for only three of the six off-axis upper-plane Te atoms included in the SOC-torque sum of Table S11, the two analyses use different Te sets. The Te atom directly above the Fe(2) vacancy lies on the vacancy axis, where the radial direction is undefined, and is excluded.

Projection	$[0, 0.10]$ eV	$[0.10, 0.20]$ eV	$[0, 0.20]$ eV
Upper Fe d	6.002340	10.314244	16.316583
Upper Te p_z	1.174756	-1.623470	-0.448714
Upper Te $p_x + p_y$	-0.169977	-0.526405	-0.696382

Table S22: Fe(2)-vacancy minus corresponding-pristine radial spin spectral weight, averaged over the selected atoms, in units of 10^{-4} states per atom, without normalization by the charge weight of the channel. Positive and negative values denote outward and inward radial components.

The upper-Te p_z difference in Table S22 is positive over $[0, 0.10]$ eV and negative over $[0, 0.20]$ eV, as is the corresponding radial vacuum polarization difference, whereas the upper-Fe d difference is positive in both intervals. Because the orbital projections neither include the propagation of the states to the tip nor partition the tunneling current, this sign correspondence is a spectral comparison only.

S8 Scope and data availability

All analyses in this Supplemental Material are based on the SCF and NSCF calculations of Sec. S1, and the input files, processed outputs, and analysis scripts are available from the authors upon reasonable request.

The calculations determine the atomic moments, their occupied orbital composition, and the differences between the two vacancy configurations, within PBE and for the single periodic cell of Sec. S1. Two limitations follow from this scope. First, PBE does not include dynamical correlation, and DFT+U calculations for bulk $Fe_{3-x}GeTe_2$ found that the simulated STM contrast of Fe(2) vacancies depends on the Hubbard U [37]. Second, the present work contains no constrained-collinear calculation with SOC and no calculation with SOC removed from individual species, and the decomposition of the SOC torque by source atom and the window-resolved spin balance do not replace such calculations. The present results therefore do not determine a canting energy, an exchange tensor, or the fraction of the surface spin signal caused by canting, and they give the net DMI only as an order-of-magnitude estimate (Sec. S6.4).

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